

A SUMMER NOTEBOOK · VOLUME I

Mathematical *Analysis* Learning Notes

on its applications, & a philosophy

essential singularity, $e^{1/z}$

Even if a mathematical framework is invented, its inventability presupposes a pre-existing space of logical possibility. What is invented is our access to the structure; what is discovered is that, once the structure is allowed, its consequences are necessary.

— X. L.

Author's Preface

Xuran Lyu



These notes are a record of learning in motion: definitions made precise, examples worked until they become familiar, and questions kept close enough to turn into structure. I am writing them as a companion for a summer of analysis, topology, measure theory, functional analysis, elliptic PDE, complex analysis, and the philosophy around mathematical practice.

The aim is not only to collect results, but to preserve the path toward them: what each idea is trying to solve, why the formal language is natural, and how different parts of mathematics quietly talk to one another. The pages that follow should therefore be read as both reference and notebook: clean where possible, exploratory where useful, and always open to revision.

My Story of Learning Mathematics. I majored in mathematics, philosophy, and economics as an undergraduate. Philosophy of mathematics, metaphysics, epistemology, and philosophy of science pushed me toward analysis because they taught me to ask what a statement really means, what counts as a rigorous argument, and how hidden assumptions shape a theory. Philosophy of science especially trained me to think more rigorously about models, evidence, explanation, and the structure of a discipline.

A philosophical instinct. My own instinct is somewhere between mathematical Platonism and structuralism. Even if a mathematical framework is invented, its inventability presupposes a pre-existing space of logical possibility. What is invented is our access to the structure; what is discovered is that, once the structure is allowed, its consequences are necessary. Here “allowed” means logically possible, or possibly true. Once the structure is fixed, its consequences are true no matter what within that structure.

I took point-set topology in Fall 2023 with Professor Nikolai Saveliev at the University of Miami, and real analysis I in Fall 2024 with Professor Ilie Grigorescu at the University of Miami. In Fall 2025 at Duke, I took complex analysis with Professor Calvin McPhail-Snyder. I also took ODE theory and applied PDE in Fall 2025, but I was overly ambitious. That experience is part of why I am writing these notes: I want a record of my reflections, mistakes, and improved approach to mathematics.

In Spring 2026, I attempted to take functional analysis with Professor Mark Stern at Duke, but I was afraid that I could end up with a horrible grade. So I decided to self-study the subject more carefully. These teachers still had an impact on me: not only through the material, but through the way they approached mathematics and the way they seemed to enjoy it.

I am currently learning measure theory, functional analysis, and PDE analysis and plan to take them in the Fall 2026 semester. I plan to apply to mathematics PhD programs and operations research PhD programs in Fall 2026. Wish me luck.

Why Analysis Feels Worth It. I believe mathematics is one of the keys to the universe, in the spirit of Roger Penrose's *The Road to Reality*. It is also a practical language for operations research, economics, quantitative trading, and quantitative research. Analysis is difficult because it takes time to adapt to its language, but I am enjoying that adaptation. The precision is part of the beauty.

Complex analysis made this especially vivid for me. Ideas such as the Riemann surface, the Riemann sphere, essential singularities, and the major theorems of the subject were simply appealing. I was also astonished to learn that contour integration is not just a beautiful formal technique: different contour methods help solve ODEs, PDEs, and even problems in financial mathematics, such as the Carr–Madan formula I met while taking a course on financial derivatives. After complex analysis, and while learning functional analysis and operator theory, quantum mechanics also began to feel more approachable. Functional analysis gave me a similar feeling with time series analysis: once functions, spaces, projections, operators, and convergence become more natural, the subject starts looking much less mysterious. If someone has no feeling for such ideas, then they have not yet found the gut-level part of mathematics.

Books and Channels That Shaped These Notes. The Rudin books have become landmarks for the path I am trying to walk:

- Baby Rudin: *Principles of Mathematical Analysis*.
- Papa Rudin: *Real and Complex Analysis*.
- Grandpa Rudin: *Functional Analysis*.

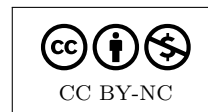
I was also influenced by math exposition online, especially [ThatMathThing](#), whose videos helped keep alive the sense that mathematics has personalities, history, and taste, not only formal statements.

Xuran Lyu

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Contents

Author's Preface	2
1 Real Analysis	6
2 Complex Analysis	8
3 Topology	10
3.1 How to Prove Something Is a Topology	11
3.2 Topological Spaces and Hausdorff Spaces	12
3.2.1 The Usual Topology	13
3.2.2 Common Types of Topologies	15
3.2.3 How to Think About Hausdorff Spaces	22
3.2.4 Why Hausdorff Spaces Matter	24
4 Measure Theory	26
5 Functional Analysis	28
5.1 How to Read This Chapter	30
5.2 Hierarchy of Spaces	31
5.3 Function and Sequence Spaces	33
5.3.1 Operator Spaces: $L(X, Y)$	34
5.4 Norms and Completeness	37
5.5 Inequalities	41
5.6 Estimate Toolkit for Continuity	47
5.7 Topological Vector Spaces	52
5.8 Operators and Functionals	54
5.8.1 Functions as Points and Functionals as Points	61
5.9 Dual Space	63
5.9.1 The Banach Space of Bounded Linear Operators	63
5.10 Integral Operators and Green's Functions	68
5.11 Hilbert Spaces and Riesz Representation	71
6 Probability Theory	73
6.1 Probability Space as a Measure Space	75
6.2 Random Variables as Measurable Maps	76
6.3 Stochastic Processes as Families of Random Variables	79
6.4 Filtrations: Information Growing in Time	81
6.5 Brownian Motion	83
6.6 Ito Integration	84
6.7 Ito Formula and Stochastic Differential Equations	86
6.8 Stochastic Volatility: Heston and Rough Heston	88
7 PDE Analysis: Elliptic PDE	93

8 Applications	95
8.1 Theory of Machine Learning	96
8.2 Reproducing Kernel Hilbert Spaces	97
8.3 Time Series Analysis	100
8.4 Quantum Mechanics	102
9 Computational Mathematics	104
9.1 Numerical Analysis	105
9.2 Numerical Linear Algebra	106
9.3 Numerical Differential Equations	107
9.4 Numerical Solvers	108
9.5 Computational Modeling	109
10 Philosophy of Math	110
11 Philosophy of Science and Quant Research	112
A Reference and Study Conventions	113
A.1 Notation	113
A.2 Learning Template	114
A.3 Question Log Convention	114
B Further Reading	115
B.1 Real Analysis	115
B.2 Topology	115
B.3 Measure Theory	115
B.4 Complex Analysis	115
B.5 Functional Analysis	115
B.6 Elliptic PDE	115
B.7 Applications and Philosophy	115

CHAPTER 1

Real Analysis

First Topics

1. Supremum and infimum; completeness of $\mathbb{R}$.
2. Sequences, subsequences, Cauchy sequences, and completeness.
3. Open and closed subsets of $\mathbb{R}^n$.
4. Continuity, uniform continuity, and compactness.
5. Differentiation in one and several variables.
6. Uniform convergence and interchange of limits.

Learning Goals. This section is for the foundations of rigorous analysis on $\mathbb{R}$ and $\mathbb{R}^n$: limits, continuity, differentiation, integration, convergence of functions, compactness, and the estimates that make later analysis possible.

Core Questions

- What does it mean for a sequence, function, or family of functions to converge?
 - Which properties survive under limits?
 - How do compactness, completeness, and continuity interact?
 - How do estimates turn intuition into proof?
-

Foundations to Strengthen.

- Practice ε - δ proofs and sequence proofs until they feel mechanical.
 - Know compactness in $\mathbb{R}^n$: Heine–Borel, Bolzano–Weierstrass, and sequential compactness.
 - Separate pointwise convergence from uniform convergence.
 - Track exactly when limits commute with continuity, derivatives, and integrals.
-

CHAPTER 2

Complex Analysis

First Topics

1. Complex numbers, geometry, and the complex plane.
2. Limits, continuity, and complex differentiability.
3. Cauchy–Riemann equations.
4. Holomorphic functions and analytic functions.
5. Contours, contour integrals, and primitives.
6. Cauchy’s theorem and Cauchy’s integral formula.
7. Power series, Laurent series, residues, and poles.
8. Conformal maps and harmonic functions.

Big Idea. Complex analysis begins like calculus on $\mathbb{C}$, but complex differentiability is rigid. Once a function is holomorphic, local behavior, power series, contour integrals, and global structure start controlling one another.

Core Questions

- What does it mean for a function $f : \mathbb{C} \rightarrow \mathbb{C}$ to be complex differentiable?
 - Why is holomorphicity much stronger than real differentiability?
 - How does integration around curves reveal information inside a domain?
 - Why are singularities classified by local series expansions?
 - How do residues compute real integrals and solve applied problems?
-

Foundations to Strengthen.

- Master Cauchy's theorem, Cauchy's integral formula, and the residue theorem as the three main engines.
 - Practice contour estimates, especially the ML inequality.
 - Classify singularities using Laurent series.
 - Connect holomorphic functions with geometry: conformal maps, harmonic functions, and analytic continuation.
 - Use residues to compute real integrals and recognize applied contour methods.
-

CHAPTER 3

Topology

First Topics

1. Topological spaces and bases.
2. Continuous maps and homeomorphisms.
3. Subspace, product, and quotient topologies.
4. Compactness and local compactness.
5. Connectedness and path connectedness.
6. Separation axioms and metric spaces.

Big Idea. Topology is the language of closeness without necessarily using a formula for distance. This section studies open sets, neighborhoods, convergence, continuity, compactness, connectedness, and quotient constructions.

Core Questions

- What structure is needed to talk about continuity?
 - How do open sets encode convergence and local behavior?
 - Why is compactness a substitute for finiteness?
 - How do different topologies on the same set change the analysis?
-

Foundations to Strengthen.

- Know bases and subbases: they are the efficient way to generate topologies.
- Read continuity as “preimages of open sets are open.”
- Understand subspace, product, and quotient topologies as three basic construction rules.
- Keep compactness, connectedness, and Hausdorffness separate.
- Build examples and counterexamples; topology is learned by seeing how definitions can fail.

Definition 3.1.

Let X be a set. A *topology* on X is a collection τ of subsets of X , called open sets, such that $\emptyset, X \in \tau$, arbitrary unions of sets in τ lie in τ , and finite intersections of sets in τ lie in τ .

Checkpoint. If two topologies live on the same set, how can one topology be “stronger” or “weaker” than the other?

3.1 How to Prove Something Is a Topology

Intuition / First Principles. To prove a collection of subsets is a topology, do not try to understand every open set one by one. Instead, check that the collection survives the three operations a topology is required to survive.

First-principles thinking. A topology is meant to encode stable local information. If many regions are allowed, then their union should also be allowed, because a point only needs to belong to one observed region. If finitely many conditions are imposed at once, their intersection should still be allowed, because we can shrink a neighborhood enough to satisfy all finitely many requirements. The empty set and whole space are the two boundary cases: observing nothing and observing everything.

Proposition 3.2: Topology Proof Strategy.

Let X be a set and let $\tau \subseteq \mathcal{P}(X)$ be a collection of subsets of X . To prove that τ is a topology on X , prove:

- (i) $\emptyset \in \tau$ and $X \in \tau$;
- (ii) if $\{U_\alpha\}_{\alpha \in A} \subseteq \tau$, then $\bigcup_{\alpha \in A} U_\alpha \in \tau$;
- (iii) if $U_1, \dots, U_n \in \tau$, then $\bigcap_{j=1}^n U_j \in \tau$.

Remark 3.3. The most common mistake is to prove closure only under finite unions. Topology requires arbitrary unions but only finite intersections. The reason is that infinitely many local restrictions may shrink away all room around a point.

Example 3.4. In the usual topology on $\mathbb{R}$, arbitrary unions are easy: if a point lies in a union, it lies in one open set from the family, so it inherits a small interval from that one set. Finite intersections use the minimum radius: if x has radius ε_1 inside U_1 and radius ε_2 inside U_2 , then it has radius $\min\{\varepsilon_1, \varepsilon_2\}$ inside $U_1 \cap U_2$. ■

Takeaway. Topology gives the general vocabulary for continuity and convergence.

May 25, 2026

3.2 Topological Spaces and Hausdorff Spaces

Learning Goals. Understand what a topological space is, why open sets are the basic data, and why the Hausdorff condition is the standard separation condition that makes limits behave like they do in metric spaces.

Intuition / First Principles. A topology tells us which subsets count as observable open regions. Once we know the open sets, we know how to talk about neighborhoods, continuity, convergence, closure, density, and compactness. A Hausdorff space is a topological space where distinct points can be separated by disjoint open neighborhoods.

First-principles thinking. Start with only a set X . A bare set has points, but it has no built-in meaning of “near,” “continuous,” or “convergent.” To do analysis, we must add exactly enough structure to say which regions are open around a point. That added structure is a topology. If we also want limits to behave like ordinary limits in $\mathbb{R}^n$, we need a way to distinguish two different points by local observations; this is the Hausdorff condition.

Definition 3.5.

A *topological space* is a pair (X, τ) , where X is a set and τ is a collection of subsets of X such that

- (i) $\emptyset \in \tau$ and $X \in \tau$;
- (ii) arbitrary unions of sets in τ are in τ ;
- (iii) finite intersections of sets in τ are in τ .

The elements of τ are called *open sets*.

Definition 3.6.

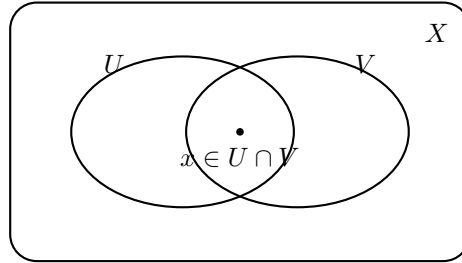
Let (X, τ) be a topological space. A set $U \subseteq X$ is a *neighborhood* of $x \in X$ if there exists an open set $O \in \tau$ such that

$$x \in O \subseteq U.$$

The important point is that topology does not begin with distance. It begins with open sets. If a metric d is available, then the open balls

$$B(x, r) = \{y \in X : d(x, y) < r\}$$

generate a topology. But many useful topologies are not introduced by first writing down a familiar distance.



Finite intersections of open sets are open: if $U, V \in \tau$, then $U \cap V \in \tau$.

Figure 1: Open sets are the primitive objects in a topology.

Example 3.7. Every metric space (X, d) is a topological space: call $U \subseteq X$ open if for every $x \in U$ there exists $r > 0$ such that $B(x, r) \subseteq U$. ■

3.2.1 The Usual Topology

Intuition / First Principles. The usual topology is the topology already used in calculus and real analysis, even before being named. On $\mathbb{R}$, it says that open intervals are the basic local regions. On $\mathbb{R}^n$, it says that open balls are the basic local regions.

First-principles thinking. To do analysis on $\mathbb{R}$, we need to say when a point has room around it. The primitive local observation is not the whole line, and not a single point; it is a small interval around the point. A set is open when every point in it has some positive amount of room that still stays inside the set.

Definition 3.8.

The *usual topology* on $\mathbb{R}$ is the topology generated by all open intervals

$$(a, b) = \{x \in \mathbb{R} : a < x < b\}.$$

Equivalently, $U \subseteq \mathbb{R}$ is open in the usual topology if for every $x \in U$ there exists $\varepsilon > 0$ such that

$$(x - \varepsilon, x + \varepsilon) \subseteq U.$$

Proposition 3.9.

The usual topology on $\mathbb{R}$ is a topology.

Proof. Let

$$\tau_{\text{usual}} = \{U \subseteq \mathbb{R} : \text{for every } x \in U, \text{ there exists } \varepsilon > 0 \text{ with } (x - \varepsilon, x + \varepsilon) \subseteq U\}.$$

We check the three topology axioms.

First, $\emptyset \in \tau_{\text{usual}}$ because there is no $x \in \emptyset$ to check. Also $\mathbb{R} \in \tau_{\text{usual}}$, since for every $x \in \mathbb{R}$ we can take any $\varepsilon > 0$, and then $(x - \varepsilon, x + \varepsilon) \subseteq \mathbb{R}$.

Second, let $\{U_\alpha\}_{\alpha \in A}$ be any family of sets in τ_{usual} . If

$$x \in \bigcup_{\alpha \in A} U_\alpha,$$

then $x \in U_{\alpha_0}$ for some $\alpha_0 \in A$. Since U_{α_0} is open, there exists $\varepsilon > 0$ such that

$$(x - \varepsilon, x + \varepsilon) \subseteq U_{\alpha_0} \subseteq \bigcup_{\alpha \in A} U_\alpha.$$

Thus arbitrary unions are open.

Third, let $U, V \in \tau_{\text{usual}}$. If $x \in U \cap V$, then there exist $\varepsilon_U, \varepsilon_V > 0$ such that

$$(x - \varepsilon_U, x + \varepsilon_U) \subseteq U, \quad (x - \varepsilon_V, x + \varepsilon_V) \subseteq V.$$

Put $\varepsilon = \min\{\varepsilon_U, \varepsilon_V\}$. Then

$$(x - \varepsilon, x + \varepsilon) \subseteq U \cap V.$$

So $U \cap V$ is open. The same argument, using the minimum of finitely many positive radii, proves that every finite intersection of usual open sets is open. Therefore τ_{usual} is a topology on $\mathbb{R}$. $\square$

Every point of an open set has a smaller interval around it.

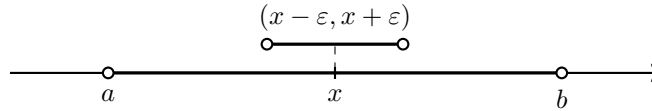


Figure 2: In the usual topology on $\mathbb{R}$, open intervals are the basic neighborhoods.

Proposition 3.10.

A subset $U \subseteq \mathbb{R}$ is open in the usual topology if and only if it is a union of open intervals.

Proof. If U is a union of open intervals, then every point $x \in U$ lies in some interval $(a, b) \subseteq U$. Choose

$$\varepsilon < \min\{x - a, b - x\}.$$

Then $(x - \varepsilon, x + \varepsilon) \subseteq (a, b) \subseteq U$, so U is open.

Conversely, if U is open, then for every $x \in U$ choose $\varepsilon_x > 0$ such that $(x - \varepsilon_x, x + \varepsilon_x) \subseteq U$. Then

$$U = \bigcup_{x \in U} (x - \varepsilon_x, x + \varepsilon_x),$$

so U is a union of open intervals. □

Example 3.11. The sets $(0, 1)$, $(0, 1) \cup (2, 5)$, $\mathbb{R}$, and $\emptyset$ are open in the usual topology on $\mathbb{R}$. ■

Example 3.12. The sets $[0, 1]$, $(0, 1]$, and $\{0\}$ are not open in the usual topology on $\mathbb{R}$. For example, $0 \in [0, 1]$, but every interval $(-\varepsilon, \varepsilon)$ around 0 contains negative numbers, so it is not contained in $[0, 1]$. ■

Definition 3.13.

The *usual topology* on $\mathbb{R}^n$ is the topology generated by Euclidean open balls

$$B(x, r) = \{y \in \mathbb{R}^n : \|x - y\|_2 < r\}, \quad r > 0.$$

Equivalently, $U \subseteq \mathbb{R}^n$ is open if every $x \in U$ has some Euclidean ball $B(x, r)$ contained in U .

Remark 3.14. In the usual topology, continuity agrees with the ε - δ definition from calculus. A function $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is continuous exactly when the preimage of every usual open set in $\mathbb{R}^m$ is usual open in $\mathbb{R}^n$.

Common Pitfall. Open does not mean “has no boundary” in an informal visual sense. Open means every point of the set has a full small neighborhood still inside the set. Also, a set can be neither open nor closed, such as $(0, 1]$ in the usual topology on $\mathbb{R}$.

Takeaway. The usual topology is the standard topology of real analysis: intervals on $\mathbb{R}$, balls on $\mathbb{R}^n$, and the familiar meaning of limits and continuity.

3.2.2 Common Types of Topologies

Intuition / First Principles. Different topologies on the same set represent different amounts of observable information. A finer topology has more open sets, so it can distinguish points and local behavior more sharply. A coarser topology has fewer open sets, so it sees less.

First-principles thinking. When a topology appears, ask: where did its open sets come from? Sometimes they come from a distance, sometimes from an order, sometimes from a larger ambient space, and sometimes from a requirement that certain maps become continuous. This question usually explains what the topology is designed to measure.

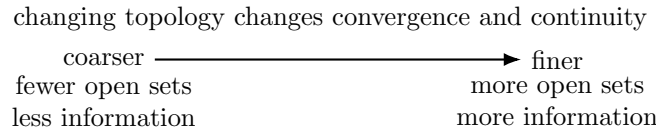


Figure 3: A topology is a choice of observational resolution.

Definition 3.15: Discrete and Indiscrete Topologies.

Let X be a set.

- The *discrete topology* is $\tau = \mathcal{P}(X)$. Every subset of X is open.
- The *indiscrete topology* is $\tau = \{\emptyset, X\}$. Only $\emptyset$ and X are open.

Remark 3.16. The discrete topology is the finest topology on X ; it has the most open sets possible. The indiscrete topology is the coarsest topology on X ; it has the fewest open sets possible.

Definition 3.17: Metric Topology.

If (X, d) is a metric space, the *metric topology* is the topology whose open sets are those $U \subseteq X$ such that for every $x \in U$ there exists $r > 0$ with

$$B(x, r) = \{y \in X : d(x, y) < r\} \subseteq U.$$

The usual topology on $\mathbb{R}^n$ is the metric topology generated by the Euclidean metric.

Definition 3.18: Cofinite Topology.

Let X be a set. The *cofinite topology* on X is

$$\tau_{\text{cofinite}} = \{\emptyset\} \cup \{U \subseteq X : X \setminus U \text{ is finite}\}.$$

So the nonempty open sets are exactly the sets whose complements are finite.

Example 3.19. On an infinite set X , the cofinite topology is much coarser than the discrete topology. It says that an open set is large if it misses only finitely many points. Any two nonempty cofinite open sets intersect, so an infinite cofinite space is not Hausdorff. ■

Definition 3.20: Subspace Topology.

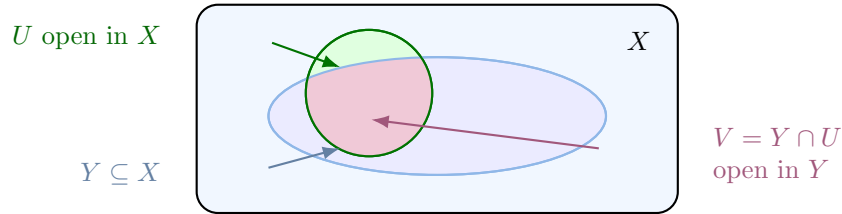
If (X, τ) is a topological space and $Y \subseteq X$, the *subspace topology* on Y is

$$\tau_Y = \{Y \cap U : U \in \tau\}.$$

Open sets in Y are obtained by cutting open sets of X down to Y . This is what it means to say that Y *inherits* a topology from X : we do not invent open sets from scratch on Y ; a set $V \subseteq Y$ is open in Y exactly when

$$V = Y \cap U$$

for some open set U in the ambient space X . Thus openness is always relative to the space being used.



Y inherits open sets by intersecting ambient open sets with Y .

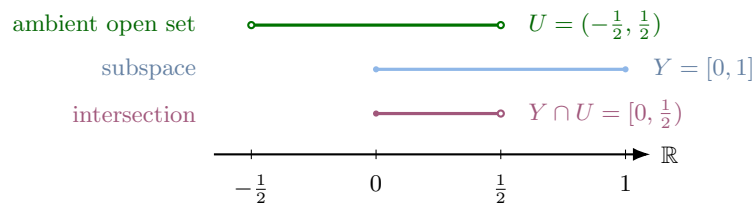
Example 3.21. The interval $[0, 1]$ inherits a topology as a subspace of $\mathbb{R}$ in the sense of Definition 3.20. The set $[0, \frac{1}{2})$ is open in $[0, 1]$ because

$$[0, \frac{1}{2}) = [0, 1] \cap (-\frac{1}{2}, \frac{1}{2}),$$

and $(-\frac{1}{2}, \frac{1}{2})$ is open in the ambient space $\mathbb{R}$. This proves openness in the subspace topology: by Definition 3.20, a set $V \subseteq [0, 1]$ is open in $[0, 1]$ if there exists an open set $U \subseteq \mathbb{R}$ such that

$$V = [0, 1] \cap U.$$

Here we take $U = (-\frac{1}{2}, \frac{1}{2})$ and get $V = [0, \frac{1}{2})$. So “open in $[0, 1]$ ” does not mean open on the whole real line. It means open after we restrict our attention to the smaller world $[0, 1]$. At the endpoint 0, we do not need room to the left inside $\mathbb{R}$; points to the left of 0 are not part of the subspace $[0, 1]$.



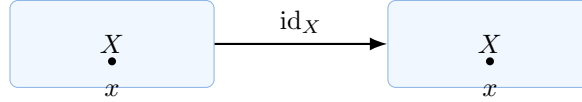
even though $[0, \frac{1}{2})$ is not open in $\mathbb{R}$ with the usual topology. ■

Definition 3.22: Identity Map.

For a set X , the *identity map* on X is the function

$$\text{id}_X : X \rightarrow X, \quad \text{id}_X(x) = x.$$

It keeps both the point and the ambient space the same.



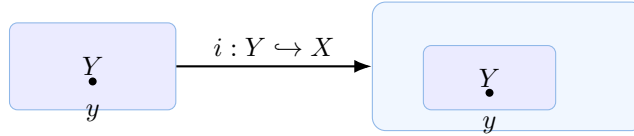
identity map: same point, same space

Definition 3.23: Inclusion Map.

If $Y \subseteq X$, the *inclusion map* is

$$i : Y \hookrightarrow X, \quad i(y) = y.$$

It does not move the point. It changes the space in which we are regarding the point: first y is viewed as a point of Y , then the same point is viewed as a point of X .

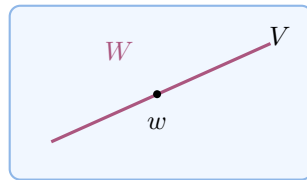
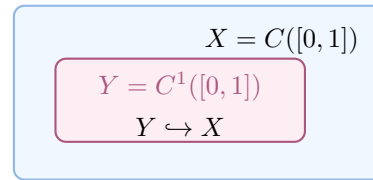


inclusion map: same point, larger ambient space

Definition 3.24: Subspaces in Linear Algebra and Functional Analysis.

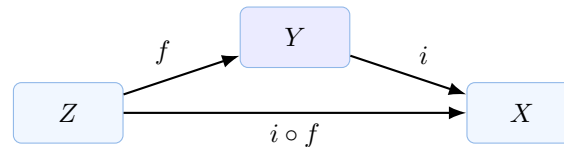
A *linear subspace* $W \subseteq V$ is a subset that is itself a vector space using the operations from V . Its natural map into V is the inclusion $W \hookrightarrow V$.

In functional analysis, the same idea appears with spaces whose points are functions. A smaller function space Y may sit inside a larger function space X , and the subspace topology says that convergence, openness, and continuity in Y are inherited from X , while staying inside Y .

linear algebra**functional analysis**

A subspace is a smaller world inside a larger one. The elements do not change; the available ambient structure changes.

Remark 3.25: Why the Subspace Topology Matters. The subspace topology matters because analysis usually happens on smaller worlds: intervals, surfaces, domains with boundary, solution sets, constraint sets, and function subspaces. It is the coarsest topology on Y that makes the inclusion map $i : Y \hookrightarrow X$ continuous. Equivalently, a map $f : Z \rightarrow Y$ is continuous exactly when $i \circ f : Z \rightarrow X$ is continuous.



continuity into the subspace can be checked after viewing the same values in the ambient space

Definition 3.26: Product Topology.

If X and Y are topological spaces, the *product topology* on $X \times Y$ is generated by products

$$U \times V,$$

where $U \subseteq X$ and $V \subseteq Y$ are open. These rectangles are the basic open sets.

Definition 3.27: Quotient Topology.

Let $q : X \rightarrow Y$ be a surjective map from a topological space X onto a set Y . The *quotient topology* on Y is defined by declaring $U \subseteq Y$ open exactly when

$$q^{-1}(U)$$

is open in X .

Remark 3.28. Subspace topology means “inherit open sets from a larger space.” Product topology means “make coordinate projections continuous.” Quotient topology means “glue points together and keep exactly the open sets whose preimages were open before gluing.”

Definition 3.29: Order Topology.

On a linearly ordered set, the *order topology* is generated by open intervals

$$(a, b) = \{x : a < x < b\},$$

together with the appropriate one-sided intervals near endpoints. The usual topology on $\mathbb{R}$ is also its order topology.

Definition 3.30: Weak Topology.

Let X be a set and let $\mathcal{F}$ be a collection of maps $f : X \rightarrow Y_f$, where each Y_f is a topological space. The *weak topology generated by $\mathcal{F}$* is the coarsest topology on X that makes every $f \in \mathcal{F}$ continuous.

Example 3.31. In functional analysis, if X is a normed vector space, the *weak topology* $\sigma(X, X^*)$ is the coarsest topology on X that makes every continuous linear functional $f \in X^*$ continuous. Thus $x_n \rightarrow x$ weakly means

$$f(x_n) \rightarrow f(x) \quad \text{for every } f \in X^*.$$

This topology is usually weaker than the norm topology. ■

Definition 3.32: Zariski Topology.

On $\mathbb{R}^n$ or $\mathbb{C}^n$, the *Zariski topology* declares algebraic sets, meaning common zero sets of polynomials, to be closed. This topology is central in algebraic geometry and is much coarser than the usual topology.

Takeaway. The common topologies to recognize first are: discrete, indiscrete, metric/usual, cofinite, subspace, product, quotient, order, weak, and Zariski. The guiding question is always: what information are the open sets designed to preserve?

Definition 3.33.

A topological space X is *Hausdorff* if for every pair of distinct points $x, y \in X$, there exist open sets $U, V \subseteq X$ such that

$$x \in U, \quad y \in V, \quad U \cap V = \emptyset.$$

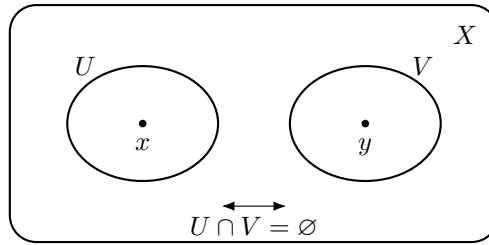


Figure 4: The Hausdorff condition separates distinct points by disjoint open neighborhoods.

Proposition 3.34.

Every metric space is Hausdorff.

Proof. Let (X, d) be a metric space and let $x \neq y$. Then $d(x, y) > 0$. Put

$$r = \frac{d(x, y)}{3}.$$

The balls $B(x, r)$ and $B(y, r)$ are disjoint. Indeed, if some z belonged to both balls, then

$$d(x, y) \leq d(x, z) + d(z, y) < r + r = \frac{2d(x, y)}{3},$$

which is impossible. Thus distinct points can be separated by disjoint open neighborhoods. $\square$

Proposition 3.35.

In a Hausdorff space, a convergent sequence has at most one limit.

Proof. Suppose $x_n \rightarrow x$ and $x_n \rightarrow y$ with $x \neq y$. Since the space is Hausdorff, choose disjoint open sets U, V with $x \in U$ and $y \in V$. Because $x_n \rightarrow x$, eventually $x_n \in U$. Because $x_n \rightarrow y$, eventually $x_n \in V$. Therefore, for all sufficiently large n , we would have $x_n \in U \cap V$, contradicting $U \cap V = \emptyset$. $\square$

3.2.3 How to Think About Hausdorff Spaces

Intuition / First Principles. The Hausdorff condition says that points are locally distinguishable. If $x \neq y$, then the topology has enough open sets to build a private open region around x and a private open region around y that do not touch.

First-principles thinking. A topology is a system of tests: open neighborhoods are the local observations we are allowed to make. If two different points can never be separated by disjoint observations, then the topology is too coarse to fully tell them apart. Hausdorffness is the minimum separation condition that restores the analytic behavior we expect from metric spaces: a limit, if it exists, should point to one place rather than two.

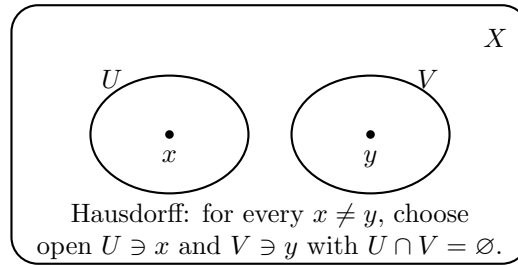


Figure 5: A Hausdorff space can separate any two distinct points.

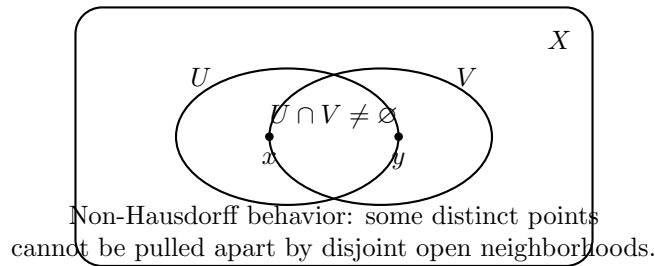


Figure 6: A non-Hausdorff space has points the topology cannot fully distinguish.

Example 3.36. The real line $\mathbb{R}$ with its usual topology is Hausdorff. If $x < y$, set

$$r = \frac{y - x}{3}.$$

Then $(x - r, x + r)$ and $(y - r, y + r)$ are disjoint open neighborhoods of x and y .

Example 3.37. Every discrete space is Hausdorff. If $x \neq y$, then $\{x\}$ and $\{y\}$ are disjoint open neighborhoods. ■

Example 3.38. If X has at least two points, the indiscrete topology $\{\emptyset, X\}$ is not Hausdorff. The only open neighborhood of any point is X itself, so two distinct points cannot have disjoint open neighborhoods. ■

Proposition 3.39.

A topological space X is Hausdorff if and only if the diagonal

$$\Delta = \{(x, x) : x \in X\}$$

is closed in the product space $X \times X$.

Proof. Assume X is Hausdorff. If $(x, y) \notin \Delta$, then $x \neq y$. Choose disjoint open sets $U, V \subseteq X$ with $x \in U$ and $y \in V$. Then $U \times V$ is an open neighborhood of (x, y) in $X \times X$, and

$$(U \times V) \cap \Delta = \emptyset.$$

Thus every point outside Δ has an open neighborhood contained in $(X \times X) \setminus \Delta$, so $(X \times X) \setminus \Delta$ is open. Therefore Δ is closed.

Conversely, assume Δ is closed. If $x \neq y$, then $(x, y) \notin \Delta$. Since $(X \times X) \setminus \Delta$ is open, there exist open sets $U, V \subseteq X$ such that

$$(x, y) \in U \times V \subseteq (X \times X) \setminus \Delta.$$

This means $x \in U$, $y \in V$, and $U \cap V = \emptyset$; otherwise a point $z \in U \cap V$ would give $(z, z) \in U \times V \cap \Delta$. Therefore X is Hausdorff. □

Remark 3.40. In very general topology, sequences do not detect all topological behavior. The fully general statement is: in a Hausdorff space, limits of nets are unique. For metric spaces and first-countable spaces, sequences are enough, so the sequence version captures the main analytic intuition.

Remark 3.41. Hausdorffness is one of the reasons topology matters in functional analysis. A topological vector space is usually assumed Hausdorff, so limits of vectors are unique. For a locally convex space, the topology is Hausdorff exactly when continuous linear functionals separate points: for every $x \neq 0$, there exists a continuous linear functional f with $f(x) \neq 0$. In particular, the weak topology $\sigma(X, X^*)$ is Hausdorff precisely when the dual space X^* separates points of X .

3.2.4 Why Hausdorff Spaces Matter

Intuition / First Principles. Hausdorffness matters because analysis depends on being able to identify where a process ends. If a sequence, net, or approximating family can converge to two different points, then the topology is too blurry for many analytic arguments.

First-principles thinking. A limit is supposed to be the final location forced by all sufficiently small observations. If two different points cannot be separated by disjoint neighborhoods, then no local observation can decisively say which point the process is approaching. Hausdorffness says: different points eventually require different local evidence.

1. **It gives uniqueness of limits.** In metric spaces, we take this for granted. In general topology it is a real theorem: Hausdorff separation prevents one convergent process from having two different limits.
2. **It makes compact sets behave like closed bounded sets.** In a Hausdorff space, compact subsets are closed. This is one of the reasons compactness becomes a powerful substitute for finiteness in analysis.
3. **It makes graphs of continuous maps closed.** If Y is Hausdorff and $f : X \rightarrow Y$ is continuous, then

$$\text{graph}(f) = \{(x, f(x)) : x \in X\}$$

is closed in $X \times Y$. This matters because closed graphs are a central idea in functional analysis.

4. **It lets functions separate information cleanly.** If X is Hausdorff, points are topologically distinguishable. In functional analysis, this becomes the question: do continuous linear functionals separate vectors?
5. **It rules out pathological identifications.** Non-Hausdorff spaces can intentionally glue points together so tightly that the topology cannot pull them apart. This can be useful in some geometry, but it is usually too weak for ordinary analysis.

Proposition 3.42.

If X is compact and Y is Hausdorff, then every continuous bijection $f : X \rightarrow Y$ is a homeomorphism.

Proof. It is enough to prove that f sends closed sets to closed sets. Let $C \subseteq X$ be closed. Since X is compact, C is compact. Since f is continuous, $f(C)$ is compact in Y . Because Y is Hausdorff, compact subsets of Y are closed. Thus $f(C)$ is closed. Therefore f is a closed map, so the inverse map $f^{-1} : Y \rightarrow X$ is continuous. $\square$

Takeaway. Hausdorffness is the topological condition that keeps distinct points distinct from the viewpoint of limits. This is why it appears so often in analysis, functional analysis, and PDE: without it, convergence may stop behaving like convergence.

Checkpoint. To test whether a space is Hausdorff, pick two arbitrary distinct points and ask: can the topology produce two disjoint open neighborhoods, one around each point?

Common Pitfall. The Hausdorff condition is not part of the definition of a topological space. Some topological spaces are not Hausdorff. In analysis, however, Hausdorff spaces are often preferred because uniqueness of limits is usually essential.

Takeaway. A topology tells us what open sets are. A Hausdorff topology is one where distinct points can be separated enough that limits are unique.

CHAPTER 4

Measure Theory

First Topics

1. Sigma-algebras and measurable spaces.
2. Measures and outer measures.
3. Lebesgue measure on $\mathbb{R}^n$.
4. Measurable functions and simple functions.
5. Lebesgue integral.
6. Monotone convergence, Fatou's lemma, and dominated convergence.
7. L^p spaces and inequalities.

Learning Goals. This section is for measurable sets, measures, measurable functions, Lebesgue integration, convergence theorems, and L^p spaces. The aim is to build the integration theory used in probability, PDE, and functional analysis.

Core Questions

- Which sets are measurable, and why can not every subset always be measured consistently?
 - What does it mean for a function to be measurable?
 - When can limits and integrals be exchanged?
 - How does measure theory create the spaces $L^p(\Omega)$?
-

Foundations to Strengthen.

- Understand σ -algebras as the events or sets a theory is allowed to measure.
 - Treat measurable functions as maps compatible with measurable sets.
 - Make monotone convergence, Fatou's lemma, and dominated convergence automatic tools.
 - Learn L^p spaces, Young's inequality, Holder's inequality, Minkowski's inequality, and completeness.
 - Study product measures, Fubini–Tonelli, and Radon–Nikodym.
-

CHAPTER 5

Functional Analysis

First Topics

1. Normed vector spaces and Banach spaces.
2. Hilbert spaces, inner products, and orthogonality.
3. Bounded linear operators.
4. Hahn–Banach theorem.
5. Uniform boundedness, open mapping, and closed graph theorems.
6. Weak and weak-* convergence.
7. Compact operators and spectral theory.

My Story. I am drawn to the philosophy of mathematics, especially mathematical Platonism, and functional analysis feels like the bridge toward that larger picture. *The Road to Reality* by Roger Penrose is one motivation. **Big Idea.** Functional analysis is linear algebra plus topology plus analysis: the linear algebra of infinite-dimensional vector spaces.

Core Questions

- What does convergence mean in a space of functions?
 - Why is completeness important?
 - Why are bounded linear maps exactly the continuous linear maps?
 - How do weak topologies and dual spaces reveal compactness?
-

Foundations to Strengthen.

- Keep a concrete example list: ℓ^p , L^p , $C([0, 1])$, Hilbert spaces, and spaces of operators.
- Know why bounded linear maps are exactly continuous linear maps.
- Practice operator norm estimates.
- Learn dual spaces through examples before the abstract theorems.
- Treat Hahn–Banach, uniform boundedness, open mapping, and closed graph as the core Banach-space theorems.
- Separate norm convergence, weak convergence, and weak-* convergence.

May 25, 2026

5.1 How to Read This Chapter

Big Idea. Functional analysis becomes organized once every example is forced to answer the same three questions:

What is the space? What is its topology or norm? What maps act on it?

The chapter should therefore move from spaces, to norms and topology, to operators and functionals.

1. First identify the *objects*: vectors, sequences, functions, or operators.
2. Then choose the *structure*: topology, metric, norm, inner product, or completeness.
3. Then study the *linear maps*: bounded operators, functionals, integral operators, and Green's operators.
4. Finally use special structure, especially Hilbert space geometry, to represent functionals and solve equations.

Takeaway. An example should not appear before the object it is an example of. The running order is: spaces first, norms next, continuity estimates next, then operators, functionals, and Green's formulas.

5.2 Hierarchy of Spaces

Big Idea. The picture is important because it shows how functional analysis is built by adding structure one layer at a time. Each inner layer carries more information than the outer layer.

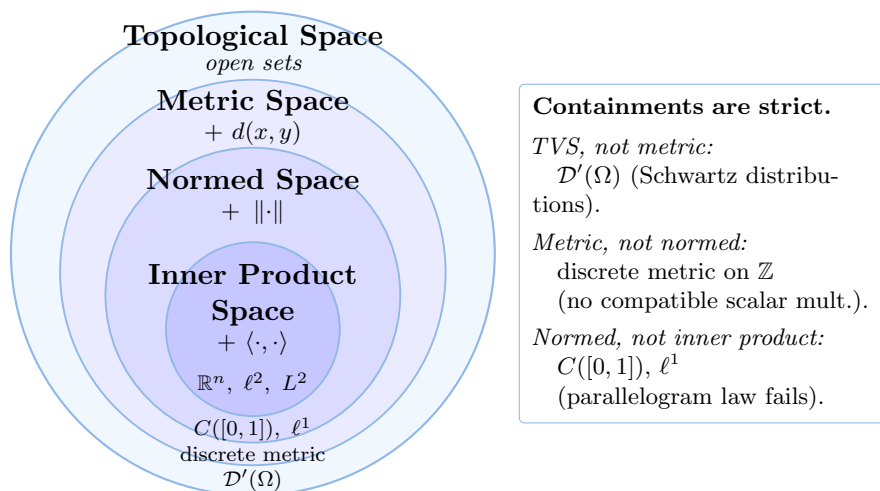


Figure 7: A hierarchy of spaces. Each ring adds the italicized structure to the one outside it. The label inside each annulus is a canonical space whose richest structure is exactly that layer, so it also witnesses that the containment is strict; the right-hand callout collects the same spaces as named counterexamples.

Intuition / First Principles. The diagram means “more structure as we move inward.” A topological space only knows open sets. A metric space knows distances. A normed vector space knows vector addition, scalar multiplication, and lengths. An inner product space knows angles as well as lengths.

First-principles thinking. Start with the weakest question: what do we need in order to talk about continuity? Open sets are enough, so we get topological spaces. If we also want numerical distances, we add a metric. If we want to add vectors and scale them while measuring their size, we add a vector space structure and a norm. If we want geometry inside the vector space, meaning angles, orthogonality, and projection, we add an inner product. Thus each step answers a stronger question than the previous step.

Proposition 5.1.

Every inner product space is a normed vector space, every normed vector space is a metric space, and every metric space is a topological space.

Proof. If V is an inner product space, define

$$\|x\| = \sqrt{\langle x, x \rangle}.$$

This is a norm, so V becomes a normed vector space. If V is a normed vector space, define

$$d(x, y) = \|x - y\|.$$

This is a metric. If (X, d) is a metric space, define a set $U \subseteq X$ to be open when for every $x \in U$ there exists $r > 0$ such that $B(x, r) \subseteq U$. These open sets form a topology. $\square$

Remark 5.2. The reverse implications are false in general. A topological space need not come from a metric. A metric space need not have any vector space structure. A normed vector space need not come from an inner product. So the diagram should be read as a hierarchy of extra structure, not as a reversible equivalence.

Example 5.3: A Normed Space That Is Not Hilbert. The space $C([0, 1])$ with the supremum norm

$$\|f\|_{\infty} = \sup_{x \in [0, 1]} |f(x)|$$

is a normed vector space, hence a metric space and a topological space. But this norm does not come from an inner product. One way to see this is that norms coming from inner products satisfy the parallelogram law,

$$\|f + g\|^2 + \|f - g\|^2 = 2\|f\|^2 + 2\|g\|^2,$$

and the supremum norm does not satisfy this identity in general. $\blacksquare$

Remark 5.4. Function spaces, sequence spaces, and operator spaces are not extra levels between “normed” and “inner product.” They are the main kinds of objects that functional analysis studies. Depending on the chosen structure, each one may be a topological vector space, a metric space, a normed space, a Banach space, or sometimes a Hilbert space.

Remark 5.5. For functional analysis, there is another important object nearby: *topological vector spaces*. A topological vector space is not shown as a simple circle inside metric spaces, because many topological vector spaces are not normed and need not be metrizable. The correct relationship is that every normed vector space is a topological vector space, but not every topological vector space comes from a norm.

Takeaway. This hierarchy is one of the main maps for analysis: topology gives continuity, metrics give distance, norms give linear size, and inner products give geometry.

5.3 Function and Sequence Spaces

Big Idea. Before talking about functionals or operators, first identify the space whose points are being studied. In functional analysis, the points are often functions, sequences, or linear maps.

Example 5.6: Function Space. The space $C([0, 1])$ is a function space. A point of this space is a continuous function

$$f : [0, 1] \rightarrow \mathbb{R}.$$

For example, $f(x) = x^2$ is a point of $C([0, 1])$.

To talk about distance, we first put structure on this set. Addition and scalar multiplication are defined pointwise:

$$(f + g)(x) = f(x) + g(x), \quad (af)(x) = af(x).$$

With these operations, $C([0, 1])$ is a vector space. Now define the supremum norm by

$$\|f\|_{\infty} = \sup_{x \in [0, 1]} |f(x)|.$$

This is a norm: it is nonnegative, it vanishes only when $f = 0$, it is homogeneous, and it satisfies the triangle inequality because

$$|(f + g)(x)| \leq |f(x)| + |g(x)|$$

for every $x \in [0, 1]$. Therefore $C([0, 1])$ is a normed vector space. The norm induces the metric

$$d(f, g) = \|f - g\|_{\infty} = \sup_{x \in [0, 1]} |f(x) - g(x)|.$$

So only after choosing this norm does it make sense to say that two functions are close in the uniform distance. ■

Example 5.7: Sequence Spaces. Typical *sequence spaces* include

$$\ell^p = \left\{ x = (x_n)_{n=1}^{\infty} : \sum_{n=1}^{\infty} |x_n|^p < \infty \right\}, \quad 1 \leq p < \infty,$$

with norm

$$\|x\|_p = \left(\sum_{n=1}^{\infty} |x_n|^p \right)^{1/p}.$$

These are normed vector spaces, and in fact Banach spaces. The special case ℓ^2 is a Hilbert space because it has the inner product

$$\langle x, y \rangle = \sum_{n=1}^{\infty} x_n \overline{y_n}.$$
■

Example 5.8: More Function Spaces. Typical *function spaces* include $C([0, 1])$, $L^p(\Omega)$, and Sobolev spaces $H^k(\Omega)$. These spaces turn functions into points of a vector space. A norm then lets us say when functions are close. For example,

$$\|f\|_\infty = \sup_{x \in [0, 1]} |f(x)|$$

on $C([0, 1])$ measures uniform closeness of functions. ■

5.3.1 Operator Spaces: $L(X, Y)$

Definition 5.9: Operator Space $L(X, Y)$.

Let X and Y be normed vector spaces over the same field $\mathbb{F}$. The operator space from X to Y is

$$L(X, Y) = \mathcal{L}(X, Y) = \{T : X \rightarrow Y : T \text{ is linear and bounded}\}.$$

Here *bounded* means that there exists a constant $C \geq 0$ such that

$$\|Tx\|_Y \leq C \|x\|_X \quad \text{for every } x \in X.$$

The norm on this operator space is the *operator norm*

$$\|T\|_{L(X, Y)} = \sup_{\|x\|_X \leq 1} \|Tx\|_Y = \sup_{x \neq 0} \frac{\|Tx\|_Y}{\|x\|_X}.$$

If $Y = \mathbb{F}$, then

$$X^* = L(X, \mathbb{F})$$

is the dual space of continuous linear functionals on X .

When no norms have been chosen, it is safer to write $\text{Lin}(X, Y)$ for the vector space of all linear maps $X \rightarrow Y$. In these notes, $L(X, Y)$ and $\mathcal{L}(X, Y)$ both mean bounded linear maps between normed spaces.

Remark 5.10: Two Notations for Operator Spaces. There is no mathematical difference between $L(X, Y)$ and $\mathcal{L}(X, Y)$ in these notes: both mean the space of bounded linear maps from X to Y . The two symbols are a notation convention.

Different books may use them differently. Some authors write $L(X, Y)$ for all linear maps and reserve $\mathcal{L}(X, Y)$ for bounded linear maps. To avoid that ambiguity, these notes use $\text{Lin}(X, Y)$ for all linear maps when boundedness is not being assumed, and use $L(X, Y) = \mathcal{L}(X, Y)$ for bounded linear operators between normed spaces.

Example 5.11: Matrix Operators. Every matrix $A \in \mathbb{F}^{m \times n}$ defines a linear map

$$T_A : \mathbb{F}^n \rightarrow \mathbb{F}^m, \quad T_A x = Ax.$$

With the usual Euclidean norms, $T_A \in L(\mathbb{F}^n, \mathbb{F}^m)$. In finite-dimensional normed spaces, every linear map is bounded, so matrix maps are the basic finite-dimensional examples of elements of $L(X, Y)$. ■

Example 5.12: Evaluation Functional. Fix $a \in [0, 1]$. On $C([0, 1])$ with the supremum norm, define

$$\delta_a : C([0, 1]) \rightarrow \mathbb{F}, \quad \delta_a(f) = f(a).$$

Then δ_a is linear, and

$$|\delta_a(f)| = |f(a)| \leq \|f\|_\infty.$$

Therefore

$$\delta_a \in L(C([0, 1]), \mathbb{F}) = C([0, 1])^*.$$

This is an operator whose input is a function and whose output is a scalar. ■

Example 5.13: Integral Operator. Define $K : C([0, 1]) \rightarrow C([0, 1])$ by

$$(Kf)(x) = \int_0^x f(t) dt.$$

The map K is linear. Also,

$$|(Kf)(x)| \leq \int_0^x |f(t)| dt \leq \|f\|_\infty \quad (0 \leq x \leq 1),$$

so

$$\|Kf\|_\infty \leq \|f\|_\infty.$$

Thus $K \in L(C([0, 1]), C([0, 1]))$. This is a typical infinite-dimensional operator: it sends one function to another function. ■

Example 5.14: A Linear Map Not in $L(X, Y)$. The derivative map

$$D : C^1([0, 1]) \rightarrow C([0, 1]), \quad Df = f',$$

is linear. But if the domain $C^1([0, 1])$ is given only the supremum norm $\|f\|_\infty$, then D is not bounded. Indeed, for

$$f_n(x) = \sin(nx),$$

we have $\|f_n\|_\infty \leq 1$, but

$$\|Df_n\|_\infty = \|n \cos(nx)\|_\infty = n.$$

So there is no single constant C with $\|Df\|_\infty \leq C\|f\|_\infty$ for all f . The lesson is that membership in $L(X, Y)$ depends on the norms placed on the domain and target spaces. ■

Remark 5.15: Function Space versus Functional. Use *function space* for a space whose points are functions. Use *space of functionals* or *dual space* for a space whose points are scalar-valued maps on another vector space. Functionals come later because they are maps *on* spaces, not the first spaces themselves.

Takeaway. Always locate the point first. In $C([0, 1])$ the point is a function; in ℓ^p the point is a sequence; in an operator space the point will be a bounded linear map, meaning a point of $L(X, Y)$.

5.4 Norms and Completeness

Intuition / First Principles. A norm is a choice of what “size” should mean. Once a norm is chosen, it automatically gives a distance by

$$d(x, y) = \|x - y\|.$$

Different norms emphasize different features: maximum size, average size, square-integrable energy, derivatives, or operator amplification.

First-principles thinking. In a vector space, subtraction tells us the error $x - y$. A norm is the rule that turns that error vector into a nonnegative number. So choosing a norm is choosing what kind of error matters.

Definition 5.16: Norm.

Let V be a vector space over $\mathbb{R}$ or $\mathbb{C}$. A *norm* on V is a map $\|\cdot\| : V \rightarrow [0, \infty)$ such that for all $x, y \in V$ and all scalars a ,

- (i) $\|x\| = 0$ if and only if $x = 0$;
- (ii) $\|ax\| = |a| \|x\|$;
- (iii) $\|x + y\| \leq \|x\| + \|y\|$.

Definition 5.17: Seminorm.

Let V be a vector space over $\mathbb{R}$ or $\mathbb{C}$. A *seminorm* on V is a map $p : V \rightarrow [0, \infty)$ such that for all $x, y \in V$ and all scalars a ,

- (i) $p(ax) = |a| p(x)$;
- (ii) $p(x + y) \leq p(x) + p(y)$.

The difference from a norm is that a seminorm is allowed to vanish on nonzero vectors. Thus

$$p(x) = 0 \quad \text{does not necessarily imply} \quad x = 0.$$

Remark 5.18: Why Seminorms Matter. A seminorm measures one feature of a vector, not necessarily the whole vector. Seminorms are useful in functional analysis because many spaces are controlled by several measurements at once: size, derivative size, local behavior, decay, or smoothness. If one wants a genuine norm, one often combines seminorms or passes to a quotient where the invisible directions have been identified with 0.

Example 5.19: Derivative Seminorm. On $C^1([0, 1])$, define

$$p(f) = \|f'\|_\infty = \sup_{x \in [0, 1]} |f'(x)|.$$

This is a seminorm because derivatives are linear:

$$(af)' = af', \quad (f + g)' = f' + g'.$$

Therefore

$$p(af) = |a|p(f), \quad p(f + g) \leq p(f) + p(g).$$

But it is not a norm. If $f(x) = 3$, then $f \neq 0$, but $f'(x) = 0$ for all x , so

$$p(f) = 0.$$

The seminorm sees variation, not vertical height. Constant functions are invisible to this measurement. ■

Definition 5.20: Banach Space.

A *Banach space* is a normed vector space that is complete in the metric induced by its norm. Completeness means every Cauchy sequence has a limit inside the same space.

Example 5.21: Euclidean Norm. On $\mathbb{R}^n$, the Euclidean norm is

$$\|x\|_2 = \left(\sum_{j=1}^n |x_j|^2 \right)^{1/2}.$$

It comes from the inner product

$$\langle x, y \rangle = \sum_{j=1}^n x_j y_j.$$

This norm measures geometric length. ■

Example 5.22: ℓ^p and ℓ^∞ Norms. For a sequence $x = (x_n)$ and $1 \leq p < \infty$, define

$$\|x\|_p = \left(\sum_{n=1}^{\infty} |x_n|^p \right)^{1/p}.$$

The limiting maximum-size norm is

$$\|x\|_\infty = \sup_{n \geq 1} |x_n|.$$

Thus ℓ^p norms measure summable size, while ℓ^∞ measures uniform boundedness of a sequence. ■

Example 5.23: L^p Norms. For a measurable function f on a measure space Ω and $1 \leq p < \infty$, the L^p norm is

$$\|f\|_{L^p} = \left(\int_{\Omega} |f(x)|^p dx \right)^{1/p}.$$

The case $p = 2$ is especially important because it comes from the inner product

$$\langle f, g \rangle = \int_{\Omega} f(x) \overline{g(x)} dx.$$

■

Remark 5.24: Why the L^p Formula Looks This Way. The pattern

$$\left(\int_{\Omega} |f(x)|^p dx \right)^{1/p}$$

has three jobs. First, $|f(x)|^p$ measures local size without cancellation. Second, the integral adds those local sizes across the whole domain, just as a sum adds the sizes of coordinates in ℓ^p . Third, the power $1/p$ returns the answer to the same scale as f .

For $p = 2$, this says

$$\|f\|_{L^2} = \left(\int_{\Omega} |f(x)|^2 dx \right)^{1/2}.$$

The square makes positive and negative values contribute positively and penalizes large values more strongly. The integral is not decorative: it is the step that collects all the small contributions from different points of the domain. Without that aggregation, we would not know the total size of the function. The square root is also necessary: it makes

$$\|cf\|_{L^2} = |c| \|f\|_{L^2},$$

whereas $\int_{\Omega} |f|^2$ alone scales like $|c|^2$ and is therefore an energy, not a norm.

Remark 5.25: Sup Norm and L^∞ Norm. On $C([0, 1])$, the supremum norm is often written like an L^∞ norm:

$$\|f\|_{\infty} = \sup_{x \in [0, 1]} |f(x)|.$$

The reason is that p -norms measure the size of a function by

$$\|f\|_p = \left(\int_0^1 |f(x)|^p dx \right)^{1/p},$$

and as $p \rightarrow \infty$, this measurement becomes controlled by the largest value of $|f|$. So L^∞ means “the limiting case where only the largest size matters.”

Remark 5.26. There is a small but important distinction. For general measurable functions,

the L^∞ norm is defined by the *essential supremum*

$$\|f\|_{L^\infty} = \operatorname{ess\,sup}_{x \in [0,1]} |f(x)|.$$

This ignores what happens on sets of measure zero, because L^p spaces identify functions that agree almost everywhere. For continuous functions on $[0, 1]$, the essential supremum agrees with the ordinary supremum, so on $C([0, 1])$ we can safely write

$$\|f\|_{L^\infty} = \|f\|_\infty.$$

Example 5.27: Why L^∞ Uses Essential Supremum. Let

$$h(x) = \begin{cases} 1, & x = 0, \\ 0, & x \neq 0. \end{cases}$$

As an ordinary bounded function, $\sup_{x \in [0,1]} |h(x)| = 1$. But $h = 0$ almost everywhere, so as an element of $L^\infty([0, 1])$,

$$\|h\|_{L^\infty} = 0.$$

This example explains why L^∞ uses essential supremum rather than ordinary supremum. Continuous functions cannot have this kind of single-point spike, which is why the two notions agree on $C([0, 1])$. ■

Example 5.28: Sobolev Norm. For PDE, a norm often needs to measure both a function and its derivatives. A typical Sobolev norm is

$$\|u\|_{H^1(\Omega)} = \left(\int_{\Omega} |u(x)|^2 \, dx + \int_{\Omega} |\nabla u(x)|^2 \, dx \right)^{1/2}.$$

This norm says that u is small only when both u and its first derivatives are small in L^2 . ■

Takeaway. The main norms to recognize are Euclidean norms, ℓ^p norms, L^p norms, supremum norms, L^∞ norms, and Sobolev norms. The operator norm appears after bounded linear operators are introduced, because it measures the size of a map rather than the size of an ordinary vector or function.

5.5 Inequalities

Big Idea. Inequalities are the grammar of estimates. They let us replace a hard quantity by simpler quantities that norms, integrals, or energy methods can control.

Theorem 5.29: Young's Inequality.

Let $p, q > 1$ satisfy

$$\frac{1}{p} + \frac{1}{q} = 1.$$

If $a, b \geq 0$, then

$$ab \leq \frac{a^p}{p} + \frac{b^q}{q}.$$

Proof. If $a = 0$ or $b = 0$, the statement is immediate. Assume $a, b > 0$. Since $\log x$ is concave on $(0, \infty)$, and since the weights $1/p$ and $1/q$ add to 1,

$$\begin{aligned} \log \left(\frac{a^p}{p} + \frac{b^q}{q} \right) &\geq \frac{1}{p} \log(a^p) + \frac{1}{q} \log(b^q) && \text{([concavity of log] 1)} \\ &= \log a + \log b && \text{([power rule for log] 2)} \\ &= \log(ab). && \text{([log of product] 3)} \end{aligned}$$

Exponentiating both sides gives

$$ab \leq \frac{a^p}{p} + \frac{b^q}{q}. \quad \text{([exponentiate (1)–(3)] 4)}$$

□

Remark 5.30: Why We Use Young's Inequality. Young's inequality is used when a proof produces a product term, but the available estimates control powers of the separate factors. It turns the mixed quantity

$$ab$$

into separate quantities

$$a^p \quad \text{and} \quad b^q.$$

This is especially useful in energy estimates: one part can often be made small and absorbed into the left side of an inequality, while the other part remains as controlled data on the right side.

Remark 5.31: The ε Version. In estimates, the most useful form is

$$ab \leq \varepsilon a^p + C_\varepsilon b^q \quad (\varepsilon > 0),$$

where C_ε is a positive constant depending on ε, p, q . For $p = q = 2$, this becomes the very common estimate

$$ab \leq \varepsilon a^2 + \frac{1}{4\varepsilon} b^2.$$

This is the version used in PDE energy estimates: the small εa^2 term can often be absorbed into the left side of an inequality, while the remaining $C_\varepsilon b^2$ term stays on the right.

Example 5.32: Splitting a Product Term. If an estimate contains a product such as $\|u\| \|f\|$, Young's inequality with $p = q = 2$ gives

$$\|u\| \|f\| \leq \varepsilon \|u\|^2 + \frac{1}{4\varepsilon} \|f\|^2.$$

So a mixed term is turned into one term involving only u and one term involving only f . ■

Theorem 5.33: Cauchy–Schwarz Inequality.

If H is an inner product space, then for all $x, y \in H$,

$$|\langle x, y \rangle| \leq \|x\| \|y\|.$$

Proof. If $y = 0$, the statement is immediate. Assume $y \neq 0$ and define

$$\lambda = \frac{\langle x, y \rangle}{\|y\|^2}. \quad ([\text{projection coefficient}] \ 1)$$

Then

$$\begin{aligned} 0 &\leq \|x - \lambda y\|^2 && ([\text{nonnegativity of norm}] \ 2) \\ &= \|x\|^2 - \frac{|\langle x, y \rangle|^2}{\|y\|^2}. && ([\text{expand using (1)}] \ 3) \end{aligned}$$

Thus

$$\begin{aligned} \frac{|\langle x, y \rangle|^2}{\|y\|^2} &\leq \|x\|^2 && ([\text{rearrange (3)}] \ 4) \\ |\langle x, y \rangle|^2 &\leq \|x\|^2 \|y\|^2 && ([\text{multiply by } \|y\|^2] \ 5) \\ |\langle x, y \rangle| &\leq \|x\| \|y\|. && ([\text{take square roots}] \ 6) \end{aligned}$$

□

Remark 5.34: Why Cauchy–Schwarz Matters. Cauchy–Schwarz says an inner product cannot be larger than the product of the two lengths. It turns a geometric interaction term $\langle x, y \rangle$ into a pure size estimate $\|x\| \|y\|$. This is the basic estimate behind projection, orthogonality, least squares, Hilbert spaces, and many energy arguments.

Example 5.35: Cauchy–Schwarz in $\mathbb{R}^n$. For $x, y \in \mathbb{R}^n$,

$$\left| \sum_{j=1}^n x_j y_j \right| = |\langle x, y \rangle| \leq \left(\sum_{j=1}^n x_j^2 \right)^{1/2} \left(\sum_{j=1}^n y_j^2 \right)^{1/2}.$$

The left side is an interaction between two vectors; the right side only uses their lengths. ■

Theorem 5.36: Holder's Inequality.

Let $1 < p, q < \infty$ satisfy

$$\frac{1}{p} + \frac{1}{q} = 1.$$

If $f \in L^p(\Omega)$ and $g \in L^q(\Omega)$, then

$$\int_{\Omega} |f(x)g(x)| \, dx \leq \|f\|_{L^p} \|g\|_{L^q}.$$

Remark 5.37: First-Principles Motivation for Holder. The problem is to control the total interaction

$$\int_{\Omega} |fg| \, dx.$$

Pointwise, the product $|f(x)g(x)|$ can be large in two ways: f can be large, or g can be large. Holder's inequality says that if f is controlled in the p -power sense and g is controlled in the dual q -power sense, then the product is integrable and its total size is controlled.

The exponents must satisfy

$$\frac{1}{p} + \frac{1}{q} = 1$$

because the proof normalizes both functions and then applies Young's inequality pointwise:

$$FG \leq \frac{F^p}{p} + \frac{G^q}{q}.$$

So Holder is not magic. It is Young's inequality applied at each point and then added up over the whole domain.

Proof. If $\|f\|_{L^p} = 0$ or $\|g\|_{L^q} = 0$, the result is immediate. Otherwise define

$$F = \frac{|f|}{\|f\|_{L^p}}, \quad G = \frac{|g|}{\|g\|_{L^q}}. \quad ([\text{normalize the two functions}] \, 1)$$

Then $\|F\|_{L^p} = 1$ and $\|G\|_{L^q} = 1$. For every $x \in \Omega$,

$$F(x)G(x) \leq \frac{F(x)^p}{p} + \frac{G(x)^q}{q}. \quad ([\text{Young's inequality pointwise}] \, 2)$$

Integrating this pointwise estimate gives

$$\int_{\Omega} FG \, dx \leq \frac{1}{p} \int_{\Omega} F^p \, dx + \frac{1}{q} \int_{\Omega} G^q \, dx \quad ([\text{integrate (2)}] \, 3)$$

$$= \frac{1}{p} + \frac{1}{q} \quad ([\text{normalization in (1)}] \, 4)$$

$$= 1. \quad ([\text{conjugate exponents}] \, 5)$$

Finally,

$$\begin{aligned} \int_{\Omega} |fg| \, dx &= \|f\|_{L^p} \|g\|_{L^q} \int_{\Omega} FG \, dx && ([\text{definition of } F, G] \, 6) \\ &\leq \|f\|_{L^p} \|g\|_{L^q} . && ([\text{use (5)}] \, 7) \end{aligned}$$

□

Remark 5.38: Holder as the L^p Cauchy–Schwarz. Holder’s inequality is the product estimate for L^p spaces. When $p = q = 2$, it becomes Cauchy–Schwarz in L^2 :

$$\int_{\Omega} |fg| \, dx \leq \|f\|_{L^2} \|g\|_{L^2} .$$

So Cauchy–Schwarz is the Hilbert-space case, while Holder is the L^p version.

Remark 5.39: Why a Measure Space Appears Here. The inequalities about L^p spaces need a measure space because L^p size is defined by adding local contributions:

$$\|f\|_{L^p} = \left(\int_{\Omega} |f|^p \, d\mu \right)^{1/p} .$$

To know what this integral means, we need a measure space $(\Omega, \mathcal{A}, \mu)$: it tells us which sets are measurable and how much weight each part of the domain has.

For finite-dimensional Cauchy–Schwarz in $\mathbb{R}^n$, no measure language is necessary; the sum over coordinates is enough. But this is secretly the same idea with counting measure:

$$\sum_{j=1}^n |x_j y_j| \leq \left(\sum_{j=1}^n |x_j|^p \right)^{1/p} \left(\sum_{j=1}^n |y_j|^q \right)^{1/q} .$$

Thus measure spaces are the general setting in which sums, integrals, probability expectations, and continuous domains are treated by one language.

Remark 5.40: Holder Points Toward Dual Spaces. Here is the formal idea, before the full discussion in [item 5.75](#). If X is a normed vector space over $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$, its *dual space* is

$$X^* = L(X, \mathbb{F}),$$

the space of bounded linear functionals $\varphi : X \rightarrow \mathbb{F}$. A functional is a rule that takes a vector as input and returns a scalar as output. Its norm is

$$\|\varphi\| = \sup_{\|x\|_X \leq 1} |\varphi(x)| .$$

Holder’s inequality creates examples of such functionals. Fix $g \in L^q(\Omega)$. Define

$$\Phi_g(f) = \int_{\Omega} f(x)g(x) \, d\mu(x) \quad (f \in L^p(\Omega)).$$

Then Φ_g is linear in f , and Holder gives

$$|\Phi_g(f)| \leq \|f\|_{L^p} \|g\|_{L^q} .$$

So Φ_g is bounded, hence

$$\Phi_g \in (L^p(\Omega))^*.$$

This is why Holder feels like a door into duality: an element of L^q can be used to measure every element of L^p by integration.

Example 5.41: Holder with $p = 3$. If $f \in L^3(\Omega)$ and $g \in L^{3/2}(\Omega)$, then

$$\int_{\Omega} |fg| \, dx \leq \|f\|_{L^3} \|g\|_{L^{3/2}}.$$

This is the same product-control idea as Cauchy–Schwarz, but the two functions are allowed to live in different L^p spaces. ■

Theorem 5.42: Minkowski Inequality.

For $1 \leq p < \infty$ and $f, g \in L^p(\Omega)$,

$$\|f + g\|_{L^p} \leq \|f\|_{L^p} + \|g\|_{L^p}.$$

Proof. For $p = 1$, integrate the pointwise triangle inequality:

$$\int_{\Omega} |f + g| \, dx \leq \int_{\Omega} |f| \, dx + \int_{\Omega} |g| \, dx. \quad ([\text{pointwise triangle inequality}] \, 1)$$

Now assume $1 < p < \infty$, and let q be the conjugate exponent. If $f + g = 0$ in L^p , there is nothing to prove. Otherwise,

$$\|f + g\|_{L^p}^p = \int_{\Omega} |f + g|^p \, dx. \quad ([\text{definition of } L^p \text{ norm}] \, 2)$$

Using $|f + g| \leq |f| + |g|$,

$$\|f + g\|_{L^p}^p \leq \int_{\Omega} (|f| + |g|) |f + g|^{p-1} \, dx. \quad ([\text{pointwise triangle inequality}] \, 3)$$

Apply Holder's inequality to each term:

$$\|f + g\|_{L^p}^p \leq \|f\|_{L^p} \left\| |f + g|^{p-1} \right\|_{L^q} + \|g\|_{L^p} \left\| |f + g|^{p-1} \right\|_{L^q} \quad ([\text{Holder on both terms}] \, 4)$$

$$= (\|f\|_{L^p} + \|g\|_{L^p}) \left\| |f + g|^{p-1} \right\|_{L^q}. \quad ([\text{factor common term}] \, 5)$$

Since $q = p/(p-1)$,

$$\left\| |f + g|^{p-1} \right\|_{L^q} = \left(\int_{\Omega} |f + g|^p \, dx \right)^{1/q} \quad ([\text{compute } (p-1)q = p] \, 6)$$

$$= \|f + g\|_{L^p}^{p-1}. \quad ([\text{because } p/q = p-1] \, 7)$$

Therefore

$$\|f + g\|_{L^p}^p \leq (\|f\|_{L^p} + \|g\|_{L^p}) \|f + g\|_{L^p}^{p-1}. \quad ([\text{combine (5) and (7)}] \ 8)$$

Dividing by $\|f + g\|_{L^p}^{p-1}$ gives

$$\|f + g\|_{L^p} \leq \|f\|_{L^p} + \|g\|_{L^p}. \quad ([\text{divide by nonzero factor}] \ 9)$$

□

Remark 5.43: Why Minkowski Matters. Minkowski is the triangle inequality for L^p spaces. It is the reason

$$\|f\|_{L^p} = \left(\int_{\Omega} |f|^p \, dx \right)^{1/p}$$

is actually a norm rather than merely a way to measure size.

Example 5.44: Minkowski for Two Functions. For $f, g \in L^2(\Omega)$,

$$\left(\int_{\Omega} |f + g|^2 \, dx \right)^{1/2} \leq \left(\int_{\Omega} |f|^2 \, dx \right)^{1/2} + \left(\int_{\Omega} |g|^2 \, dx \right)^{1/2}.$$

So the L^2 distance behaves like ordinary length: the size of a sum is at most the sum of the sizes.

■

Takeaway. Young splits products, Cauchy–Schwarz controls inner products, Holder controls products in L^p , and Minkowski gives the triangle inequality for L^p norms.

5.6 Estimate Toolkit for Continuity

May 26, 2026

Big Idea. Continuity in normed spaces is usually proved by estimates. The same few inequalities explain why norms are continuous, why addition is continuous, why scalar multiplication is continuous, and why bounded linear maps are continuous.

Remark 5.45: Common Triangle Inequality Patterns. The same idea appears in several forms.

$$|a + b| \leq |a| + |b| \quad \text{for real or complex numbers,}$$

$$\|u + v\| \leq \|u\| + \|v\| \quad \text{for vectors in a normed space,}$$

and

$$d(x, z) \leq d(x, y) + d(y, z) \quad \text{for points in a metric space.}$$

The interpretation is always the same: going directly is no longer than going by way of an intermediate object.

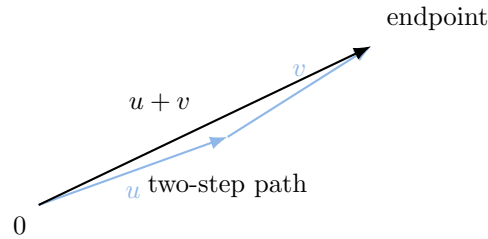


Figure 8: The vector triangle inequality says the direct displacement $u + v$ has length at most the two-step path with lengths $\|u\|$ and $\|v\|$.

Definition 5.46: Lipschitz Map.

Let (X, d_X) and (Y, d_Y) be metric spaces. A map $f : X \rightarrow Y$ is *Lipschitz* if there exists a constant $L \geq 0$ such that

$$d_Y(f(x), f(y)) \leq L d_X(x, y) \quad \text{for all } x, y \in X.$$

The number L is a Lipschitz constant. Informally, f can stretch distances by at most the fixed factor L .

Proposition 5.47: Lipschitz Implies Continuous.

Every Lipschitz map is continuous.

Proof. Let $f : X \rightarrow Y$ be Lipschitz with constant L . If $L = 0$, then f is constant and therefore

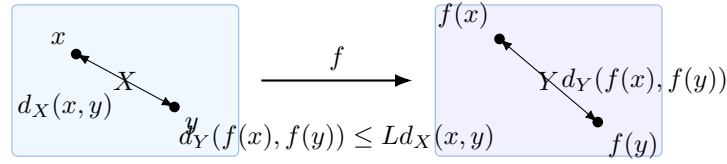


Figure 9: A Lipschitz map has controlled stretching: close inputs remain close outputs, with one fixed constant controlling all pairs.

continuous. If $L > 0$, given $\varepsilon > 0$, choose

$$\delta = \frac{\varepsilon}{L}.$$

Whenever $d_X(x, y) < \delta$,

$$d_Y(f(x), f(y)) \leq L d_X(x, y) < L \delta = \varepsilon.$$

This is exactly the ε - δ definition of continuity. □

Example 5.48: Examples of Lipschitz Maps. The map $f : \mathbb{R} \rightarrow \mathbb{R}$ given by

$$f(x) = 3x - 2$$

is Lipschitz with constant $L = 3$, because

$$|f(x) - f(y)| = |3x - 3y| = 3|x - y|.$$

The sine function is Lipschitz with constant $L = 1$:

$$|\sin x - \sin y| \leq |x - y|.$$

This follows from the mean value theorem, since

$$|(\sin)'(t)| = |\cos t| \leq 1.$$

In any normed space, the norm map

$$x \mapsto \|x\|$$

is Lipschitz with constant $L = 1$, because the reverse triangle inequality gives

$$||x| - |y|| \leq \|x - y\|.$$

The map $f(x) = x^2$ is not Lipschitz on all of $\mathbb{R}$, because its slope becomes arbitrarily large. But it is Lipschitz on every bounded interval such as $[-R, R]$, since

$$|x^2 - y^2| = |x - y| |x + y| \leq 2R |x - y|$$

whenever $x, y \in [-R, R]$. ■

Remark 5.49: How to Choose δ in an ε - δ Proof. The goal is always to force

$$d_Y(f(x), f(a)) < \varepsilon$$

from the assumption

$$d_X(x, a) < \delta.$$

The practical method is to estimate the output error in terms of the input error, then choose δ so that the estimate becomes less than ε .

For a Lipschitz map,

$$d_Y(f(x), f(a)) \leq L d_X(x, a).$$

Thus it is enough to make

$$L d_X(x, a) < \varepsilon.$$

So choose

$$\delta = \frac{\varepsilon}{L} \quad (L > 0).$$

This is the cleanest possible case: Lipschitz estimates give δ immediately.

Example 5.50: A Non-Lipschitz Delta Choice: x^2 . Prove that $f(x) = x^2$ is continuous at a fixed point $a \in \mathbb{R}$.

Solution. The goal is to show that for every $\varepsilon > 0$ there exists $\delta > 0$ such that

$$|x - a| < \delta \implies |x^2 - a^2| < \varepsilon.$$

Factor the output error:

$$|x^2 - a^2| = |(x - a)(x + a)| = |x - a| |x + a|. \quad ([\text{difference of squares}] \ 1)$$

The input error $|x - a|$ is visible, but the extra factor $|x + a|$ must be controlled.

Localize x near a by first imposing the preliminary restriction

$$|x - a| < 1. \quad ([\text{local restriction}] \ 2)$$

Then

$$\begin{aligned} |x| &= |(x - a) + a| && ([\text{rewrite } x \text{ near } a] \ 3) \\ &\leq |x - a| + |a| && ([\text{triangle inequality}] \ 4) \\ &< 1 + |a|. && ([\text{use (2)}] \ 5) \end{aligned}$$

Bound the extra factor:

$$\begin{aligned} |x + a| &\leq |x| + |a| && ([\text{triangle inequality}] \ 6) \\ &< 1 + 2|a|. && ([\text{use (5)}] \ 7) \end{aligned}$$

Return to the output error. Combining (1) and (7),

$$\begin{aligned} |x^2 - a^2| &= |x - a| |x + a| && ([\text{from (1)}] \ 8) \\ &< (1 + 2|a|) |x - a|. && ([\text{use (7)}] \ 9) \end{aligned}$$

Therefore, to make $|x^2 - a^2| < \varepsilon$, it is enough to require

$$(1 + 2|a|)|x - a| < \varepsilon. \quad ([\text{target inequality}] \ 10)$$

This is guaranteed by

$$|x - a| < \frac{\varepsilon}{1 + 2|a|}. \quad ([\text{solve for input error}] \ 11)$$

The two restrictions are (2) and (11), so choose

$$\delta = \min \left\{ 1, \frac{\varepsilon}{1 + 2|a|} \right\}. \quad ([\text{satisfy both restrictions}] \ 12)$$

Then $|x - a| < \delta$ implies both $|x - a| < 1$ and $|x - a| < \varepsilon/(1 + 2|a|)$. Hence (9) gives

$$|x^2 - a^2| < \varepsilon.$$

Thus x^2 is continuous at a . ■

Remark 5.51: Why the Triangle Inequality Gives Continuity. The triangle inequality does not say “continuity” by itself. It gives an estimate that turns small input errors into a small output error.

For addition, compare two nearby input pairs (x_1, y_1) and (x_2, y_2) . The output error is

$$\|(x_1 + y_1) - (x_2 + y_2)\| = \|(x_1 - x_2) + (y_1 - y_2)\|.$$

By the triangle inequality,

$$\|(x_1 + y_1) - (x_2 + y_2)\| \leq \|x_1 - x_2\| + \|y_1 - y_2\|.$$

This is the same pattern as the familiar real-number inequality

$$|a + b| \leq |a| + |b|.$$

The only change is that the objects being added are vectors. Set

$$u = x_1 - x_2, \quad v = y_1 - y_2.$$

Then

$$(x_1 + y_1) - (x_2 + y_2) = u + v,$$

so the normed-space triangle inequality gives

$$\|u + v\| \leq \|u\| + \|v\|.$$

Thus if x_1 is close to x_2 and y_1 is close to y_2 , then the sum $x_1 + y_1$ is close to the sum $x_2 + y_2$. In fact, with the product metric

$$D((x_1, y_1), (x_2, y_2)) = \|x_1 - x_2\| + \|y_1 - y_2\|,$$

addition is 1-Lipschitz:

$$\|(x_1 + y_1) - (x_2 + y_2)\| \leq D((x_1, y_1), (x_2, y_2)).$$

By [item 5.47](#), addition is continuous.

Remark 5.52: Scalar Multiplication. Scalar multiplication has one extra issue: both the scalar and the vector can move. The useful decomposition is

$$bx - ay = b(x - y) + (b - a)y.$$

Therefore

$$\|bx - ay\| \leq |b| \|x - y\| + |b - a| \|y\|.$$

If (b, x) is close to (a, y) , then b is close to a , so b stays bounded, say $|b| \leq |a| + 1$ near a . Hence

$$\|bx - ay\| \leq (|a| + 1) \|x - y\| + |b - a| \|y\|,$$

and the right-hand side goes to 0 as $b \rightarrow a$ and $x \rightarrow y$. This is exactly continuity of the map

$$\mathbb{F} \times V \rightarrow V, \quad (a, x) \mapsto ax.$$

Takeaway. Delta choices and continuity proofs usually come from working backward. Estimate the output error, identify what must be made small, and then choose the input restriction that forces it.

5.7 Topological Vector Spaces

Learning Goals. Understand topological vector spaces as vector spaces whose topology is compatible with addition and scalar multiplication. This is the common general setting behind normed spaces, Banach spaces, Hilbert spaces, weak topologies, and many spaces of distributions.

Intuition / First Principles. A topological vector space is a vector space where it makes sense to say that vectors are close, and where the vector space operations respect that notion of closeness. If x_n is close to x and y_n is close to y , then $x_n + y_n$ should be close to $x + y$. If scalars a_n are close to a and vectors x_n are close to x , then $a_n x_n$ should be close to ax .

First-principles thinking. Start with a vector space V . Linear algebra gives two operations: addition and scalar multiplication. But analysis also needs limits, so we add a topology. The topology is useful only if it agrees with the algebra: small changes in inputs should create small changes in sums and scalar multiples. Therefore the definition is forced by requiring the two structure maps

$$(x, y) \mapsto x + y \quad \text{and} \quad (a, x) \mapsto ax$$

to be continuous.

Definition 5.53: Topological Vector Space.

Let V be a vector space over $\mathbb{F}$, where $\mathbb{F}$ is either $\mathbb{R}$ or $\mathbb{C}$. A *topological vector space* is a vector space V together with a topology such that the maps

$$V \times V \rightarrow V, \quad (x, y) \mapsto x + y,$$

and

$$\mathbb{F} \times V \rightarrow V, \quad (a, x) \mapsto ax,$$

are continuous.

The first condition says addition is continuous. The second says scalar multiplication is continuous. Together, they mean that the topology and the linear structure are not fighting each other.

$$\begin{array}{ccc} V \times V & \xrightarrow{+} & V \\ (x, y) & & x + y \end{array} \qquad \begin{array}{ccc} \mathbb{F} \times V & \xrightarrow{\text{scalar multiplication}} & V \\ (a, x) & & ax \end{array}$$

Figure 10: A topological vector space requires both structure maps to be continuous.

Example 5.54: Normed Spaces are Topological Vector Spaces. Every normed vector space is a topological vector space. If V is a normed space, the norm defines a metric

$$d(x, y) = \|x - y\|.$$

Addition is continuous because

$$\|(x_1 + y_1) - (x_2 + y_2)\| \leq \|x_1 - x_2\| + \|y_1 - y_2\|.$$

Scalar multiplication is continuous because

$$\|a_n x_n - ax\| \leq |a_n| \|x_n - x\| + |a_n - a| \|x\|,$$

so if $a_n \rightarrow a$ and $x_n \rightarrow x$, then $a_n x_n \rightarrow ax$. For the estimate-based explanation of why these inequalities imply continuity, see [items 5.51](#) and [5.52](#). ■

Example 5.55: Weak Topology. The weak topology on a normed space X is the topology in which $x_n \rightharpoonup x$ means

$$\varphi(x_n) \rightarrow \varphi(x) \quad \text{for every } \varphi \in X^*.$$

This topology is usually weaker than the norm topology, but it is still compatible with vector addition and scalar multiplication. Thus it is a topological vector space topology. ■

Remark 5.56. Many authors include Hausdorffness in the definition of a topological vector space, while others state it separately. In a Hausdorff topological vector space, limits are unique. This is usually the setting wanted in functional analysis.

Big Idea. In a topological vector space, the behavior near every point is controlled by the behavior near 0. This happens because translation $x \mapsto x + a$ is continuous with continuous inverse $x \mapsto x - a$. So neighborhoods of 0 are the basic local objects.

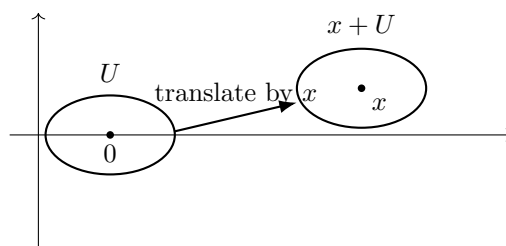


Figure 11: Neighborhoods of any point are translates of neighborhoods of 0.

Checkpoint. Why is a normed vector space automatically a topological vector space?

Solution. The norm gives a metric, hence a topology. The triangle inequality shows that addition is continuous, and the norm estimate for $a_n x_n - ax$ shows scalar multiplication is continuous. □

Takeaway. Topological vector spaces are the natural meeting point of topology and linear algebra. Normed spaces are the most familiar examples, but weak topologies show why functional analysis needs this more general language.

5.8 Operators and Functionals

Big Idea. Once the spaces are clear, the next objects are maps between them. Operators send vectors or functions to other vectors or functions. Functionals are the special case where the output is a scalar.

Definition 5.57: Bounded Linear Operator.

Let X and Y be normed spaces. A linear map $T : X \rightarrow Y$ is *bounded* if there exists $C \geq 0$ such that

$$\|Tx\|_Y \leq C \|x\|_X \quad \text{for every } x \in X.$$

The least possible such constant is the *operator norm*

$$\|T\| = \sup_{\|x\|_X \leq 1} \|Tx\|_Y.$$

Remark 5.58: Equivalent Ways to Read the Operator Norm. For a bounded linear operator $T : X \rightarrow Y$, the operator norm can be read in three equivalent ways:

$$\|T\| = \sup_{\|x\|_X \leq 1} \|Tx\|_Y = \sup_{\|x\|_X = 1} \|Tx\|_Y = \sup_{x \neq 0} \frac{\|Tx\|_Y}{\|x\|_X}.$$

The last expression is often the most intuitive: it asks for the largest possible amplification ratio

$$\frac{\text{output size}}{\text{input size}} = \frac{\|Tx\|_Y}{\|x\|_X}.$$

Why is this the same as only checking unit vectors? If $x \neq 0$, write

$$u = \frac{x}{\|x\|_X}.$$

Then $\|u\|_X = 1$ and

$$x = \|x\|_X u.$$

By linearity,

$$Tx = T(\|x\|_X u) = \|x\|_X Tu,$$

so

$$\frac{\|Tx\|_Y}{\|x\|_X} = \|Tu\|_Y.$$

Thus every ratio for a nonzero vector x is already achieved by looking at the normalized direction $u = x / \|x\|_X$.

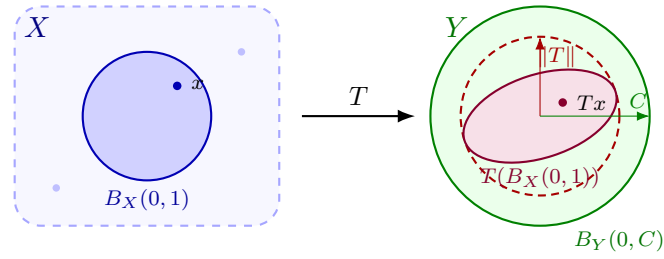


Figure 12: Schematic picture of boundedness. Because T is linear, the image $T(B_X(0, 1))$ is an ellipse (purple), not an arbitrary blob. The smallest closed ball that contains it has radius $\|T\|$ (dashed red), so the ellipse is tangent to that circle. Any larger $C \geq \|T\|$ also works as a bound, so the green ball $B_Y(0, C)$ trivially contains the image; the operator norm $\|T\|$ is the infimum over all admissible C .

Definition 5.59: Bounded Linear Functional.

Let X be a normed vector space over $\mathbb{F}$. A *linear functional* on X is a linear map

$$\varphi : X \rightarrow \mathbb{F}.$$

It is a *bounded linear functional* if there exists $C \geq 0$ such that

$$|\varphi(x)| \leq C \|x\|_X \quad \text{for every } x \in X.$$

Thus a bounded linear functional is just a bounded linear operator whose target space is the scalar field $\mathbb{F}$.

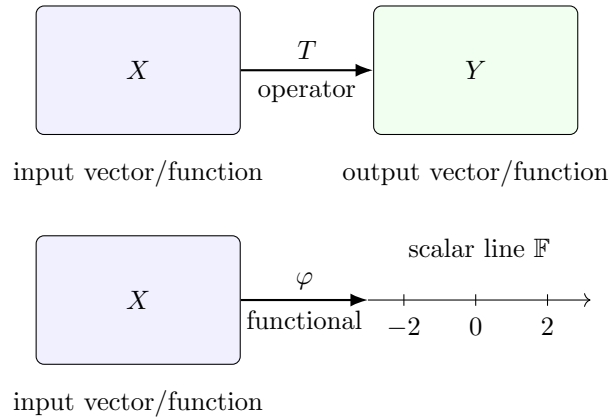
Remark 5.60: Operator Versus Functional. The difference is the output.

$$T : X \rightarrow Y \quad \text{operator: output lives in another normed space } Y,$$

while

$$\varphi : X \rightarrow \mathbb{F} \quad \text{functional: output is one scalar.}$$

So every bounded linear functional is a bounded linear operator, but not every bounded linear operator is a functional. For example, $T(f) = f'$ is an operator when its output is another function, while $\varphi(f) = f(a)$ is a functional when its output is the number $f(a)$.



Visual. An operator keeps vector-valued information. A functional collapses the input to one scalar measurement.

Theorem 5.61: Bounded if and only if Continuous.

Let X and Y be normed spaces and let $T : X \rightarrow Y$ be linear. The following three statements are equivalent:

(A) T is bounded: there exists $M \geq 0$ such that

$$\|Tx\|_Y \leq M \|x\|_X \quad \text{for every } x \in X.$$

(B) T is continuous on all of X .

(C) T is continuous at 0.

Remark 5.62: How to Read This Theorem. For a general function, continuity at one point says almost nothing about continuity elsewhere. For a *linear* map, continuity at 0 is much stronger. If we can control T near 0, then any large vector x can be scaled down into that small neighborhood. Linearity lets us scale the output back up. That scaling argument produces one global constant C such that

$$\|Tx\|_Y \leq C \|x\|_X \quad \text{for every } x \in X.$$

That is exactly boundedness.

Remark 5.63: Why Choose $\varepsilon = 1$? In the direction “continuous at 0 implies bounded,” we do not need to prove a new limit statement. We only need one finite output bound near 0. So we apply continuity at 0 with the convenient tolerance $\varepsilon = 1$. Continuity gives a radius $\delta > 0$ such that

$$\|u\|_X < \delta \quad \implies \quad \|Tu\|_Y < 1.$$

The number 1 is not special; any fixed positive tolerance would work. The radius δ is not arbitrary: it is some radius that continuity guarantees works. If this δ works, then every smaller positive radius also works.

Remark 5.64: Why Define $u = \frac{\delta}{2\|x\|}x$? Fix a nonzero vector x . The vector

$$\frac{x}{\|x\|_X}$$

is the normalized version of x : it points in the same direction as x , but has norm 1. Multiplying it by $\delta/2$ gives

$$u = \frac{\delta}{2} \frac{x}{\|x\|_X} = \frac{\delta}{2\|x\|_X} x.$$

So u is exactly a small vector in the direction of x , with size

$$\|u\|_X = \frac{\delta}{2} < \delta.$$

The factor $1/2$ has no deep meaning. It only puts u safely inside the open ball $\|u\|_X < \delta$. We could use $\delta/3$ or 0.99δ instead. We avoid using δ itself because that would give $\|u\|_X = \delta$, while the continuity statement asks for strict inequality $\|u\|_X < \delta$.

Remark 5.65: What Does “Prove Continuity” Mean Here?. To prove that T is continuous at a point a , the task is:

given any allowed output error $\varepsilon > 0$, find an input radius $\delta > 0$

such that

$$\|x - a\|_X < \delta \quad \implies \quad \|Tx - Ta\|_Y < \varepsilon.$$

So continuity is an error-control statement:

$$\text{small input error} \quad \implies \quad \text{small output error}.$$

For a linear map, the output error has a special form:

$$Tx - Ta = T(x - a).$$

If T is bounded, then

$$\|Tx - Ta\|_Y = \|T(x - a)\|_Y \leq M \|x - a\|_X.$$

Thus boundedness gives a machine for choosing δ : if we want $M \|x - a\|_X < \varepsilon$, it is enough to require

$$\|x - a\|_X < \frac{\varepsilon}{M}.$$

That is the whole idea of $(A) \Rightarrow (B)$.

Proof. Step 1: (A) implies (B). Assume T is bounded. Then there is $M \geq 0$ such that

$$\|Tx\|_Y \leq M \|x\|_X \quad \text{for every } x \in X.$$

We prove continuity at an arbitrary point $a \in X$.

Continuity target. Given any $\varepsilon > 0$, we must find $\delta > 0$ such that

$$\|x - a\|_X < \delta \quad \implies \quad \|Tx - Ta\|_Y < \varepsilon.$$

If $M = 0$, then $T = 0$, so every output error is 0 and continuity is immediate. Now assume $M > 0$ and choose

$$\delta = \frac{\varepsilon}{M}.$$

Whenever $\|x - a\|_X < \delta$,

$$\begin{aligned} \|Tx - Ta\|_Y &= \|T(x - a)\|_Y && \text{([turn output error into input error] 1)} \\ &\leq M \|x - a\|_X && \text{([boundedness controls the error] 2)} \\ &< M\delta && \text{([because } \|x - a\|_X < \delta \text{] 3)} \\ &= \varepsilon. && \text{([choice of } \delta \text{] 4)} \end{aligned}$$

Thus T is continuous at a . Since a was arbitrary, T is continuous on all of X .

Step 2: (B) implies (C). If T is continuous on all of X , then it is continuous at the particular point 0.

Step 3: (C) implies (A). Assume T is continuous at 0. Apply continuity at 0 with output tolerance $\varepsilon = 1$. Then there exists $\delta > 0$ such that

$$\|u\|_X < \delta \quad \implies \quad \|Tu\|_Y < 1.$$

Fix $x \neq 0$ and define

$$u = \frac{\delta}{2} \frac{x}{\|x\|_X} = \frac{\delta}{2} \frac{x}{\|x\|_X}.$$

This is just the normalized direction of x , scaled down to length $\delta/2$. Then

$$\begin{aligned} \|u\|_X &= \frac{\delta}{2\|x\|_X} \|x\|_X && \text{([definition of } u \text{] 5)} \\ &= \frac{\delta}{2} && \text{([cancel the input size] 6)} \\ &< \delta. && \text{([strictly inside the continuity ball] 7)} \end{aligned}$$

Therefore $\|Tu\|_Y < 1$. Using linearity again,

$$\begin{aligned} \frac{\delta}{2\|x\|_X} \|Tx\|_Y &= \left\| \frac{\delta}{2\|x\|_X} Tx \right\|_Y && \text{([homogeneity of the norm] 8)} \\ &= \left\| T \left(\frac{\delta}{2\|x\|_X} x \right) \right\|_Y && \text{([linearity] 9)} \\ &= \|Tu\|_Y && \text{([definition of } u \text{] 10)} \\ &< 1. && \text{([continuity at 0] 11)} \end{aligned}$$

Hence

$$\|Tx\|_Y < \frac{2}{\delta} \|x\|_X.$$

The same estimate is trivial when $x = 0$. Thus T is bounded with constant $M = 2/\delta$. This proves (A). $\square$

Remark 5.66: How Would You Find This Proof Yourself? This proof should not feel like a random trick. The way to find it is to write the *target* and the *available tool* next to each other.

Target. To prove that T is bounded, we need one constant M such that

$$\|Tx\|_Y \leq M \|x\|_X \quad \text{for every } x \in X.$$

So we must control $\|Tx\|_Y$ for an arbitrary large input x .

Available tool. Continuity at 0 only gives local information: using $\varepsilon = 1$, there exists $\delta > 0$ such that

$$\|u\|_X < \delta \quad \implies \quad \|Tu\|_Y < 1.$$

This only controls inputs u near 0, not arbitrary large x .

Bridge. Because T is linear, we are allowed to scale inputs. Given a large nonzero x , shrink it into the local ball:

$$u = \frac{\delta}{2\|x\|_X} x.$$

Then $\|u\|_X = \delta/2 < \delta$, so the available tool says

$$\|Tu\|_Y < 1.$$

But u is a scalar multiple of x , so linearity gives

$$Tu = T\left(\frac{\delta}{2\|x\|_X} x\right) = \frac{\delta}{2\|x\|_X} Tx.$$

Therefore

$$\frac{\delta}{2\|x\|_X} \|Tx\|_Y < 1,$$

which rearranges to

$$\|Tx\|_Y < \frac{2}{\delta} \|x\|_X.$$

Now the bound has appeared naturally: $M = 2/\delta$.

So the proof-discovery recipe is:

write the target
 $\Downarrow$
 write what the hypothesis gives
 $\Downarrow$
 build a bridge between them.

Here the bridge is normalization plus linearity.

Remark 5.67: Which “Bounded” Is This? Here *bounded* does not mean that an input function is pointwise bounded, and it does not mean that the input belongs to L^2 or L^p . Those are different ideas.

For a function f , pointwise bounded means

$$\sup_x |f(x)| < \infty.$$

For an L^p function, integrability means

$$\int |f|^p < \infty.$$

For an operator $T : X \rightarrow Y$, bounded means

$$\|Tx\|_Y \leq C \|x\|_X \quad \text{for every input } x.$$

So a bounded linear operator may act on L^2 or L^p spaces, but boundedness is a property of the rule T , not a statement that T is an L^2 or L^p function.

Remark 5.68: Why Bounded and Linear Belong Together. Linearity gives algebra:

$$T(x + y) = Tx + Ty, \quad T(\alpha x) = \alpha Tx.$$

Boundedness gives analysis:

$$\|Tx\|_Y \leq C \|x\|_X.$$

Together they say that T respects vector-space structure and also respects limits. This is why bounded linear maps are the correct morphisms in normed spaces: they are exactly the linear maps that are continuous. Without boundedness, a linear map can destroy convergence; with boundedness, estimates, limits, Cauchy sequences, dual spaces, and operator spaces become manageable.

Example 5.69: Operator Norm. If $T : X \rightarrow Y$ is a bounded linear operator between normed spaces, its operator norm

$$\|T\| = \sup_{\|x\|_X \leq 1} \|Tx\|_Y$$

measures the largest amount by which T can stretch a vector of size at most 1. ■

Example 5.70: Operator Space Revisited. By [item 5.9](#), if X and Y are normed spaces, the space

$$L(X, Y) = \mathcal{L}(X, Y)$$

of bounded linear operators $T : X \rightarrow Y$ is an operator space. It has the operator norm

$$\|T\| = \sup_{\|x\|_X \leq 1} \|Tx\|_Y.$$

The dual space $X^* = L(X, \mathbb{F})$ is the operator space of continuous linear functionals on X . ■

5.8.1 Functions as Points and Functionals as Points

Intuition / First Principles. The clean distinction is this: in a function space, a point is a function. In a space of functionals, a point is a rule that eats a function or vector and returns a scalar.

First-principles thinking. A mathematical object can play different roles depending on the space we put it in. A function $g \in L^2(\Omega)$ can be a point of the Hilbert space $L^2(\Omega)$. The same g can also generate a functional

$$F_g(f) = \int_{\Omega} f(x) \overline{g(x)} dx.$$

But g and F_g are not literally the same kind of object: g is an input vector, while F_g is a scalar-valued map on vectors.

Example 5.71: Space of Functionals. The dual space $C([0, 1])^*$ is a space whose points are continuous linear functionals on $C([0, 1])$. For example, for a fixed $a \in [0, 1]$, the evaluation map

$$\delta_a : C([0, 1]) \rightarrow \mathbb{R}, \quad \delta_a(f) = f(a),$$

is a point of $C([0, 1])^*$. It is not a function on $[0, 1]$; it is a function whose input is another function. ■

Example 5.72: Integral Functional. If $g \in C([0, 1])$, then

$$I_g(f) = \int_0^1 f(x)g(x) dx$$

defines a continuous linear functional $I_g \in C([0, 1])^*$. Here g is a point of the function space $C([0, 1])$, while I_g is a point of the functional space $C([0, 1])^*$. ■

Example 5.73: Examples of Integral Functionals. Here are common scalar-valued functionals on a function space:

$$A(f) = \int_0^1 f(t) dt \quad \text{average or total mass,}$$

$$M_1(f) = \int_0^1 tf(t) dt \quad \text{first moment,}$$

and, more generally,

$$I_g(f) = \int_0^1 f(t)g(t) dt.$$

In all three examples, the input is a function f , but the output is one number. That is why they are functionals. ■

Remark 5.74: Delta as an Evaluation Functional. The evaluation functional

$$\delta_a(f) = f(a)$$

is not an ordinary integral against a continuous function g . But it can be written formally as

$$\delta_a(f) = \int_0^1 f(t) \delta_a(t) dt,$$

where δ_a is the Dirac delta distribution concentrated at a . More precisely, it is integration against the Dirac measure:

$$\delta_a(f) = \int_{[0,1]} f(t) d\delta_a(t).$$

So the delta symbol is not an ordinary function. It is a way of encoding the functional “evaluate f at a .”

Takeaway. The output decides the name. A rule with scalar output is a functional; a rule with vector or function output is an operator.

5.9 Dual Space

Big Idea. The dual space collects all continuous linear measurements of a normed space. Before specializing to scalar-valued maps, it is better to prove the general fact: if the target space is complete, then the whole space of bounded linear operators is complete.

Definition 5.75: Dual Space.

Let X be a normed vector space over $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$. The *dual space* of X is

$$X^* = L(X, \mathbb{F}),$$

the normed space of all bounded linear functionals $\varphi : X \rightarrow \mathbb{F}$ with norm

$$\|\varphi\| = \sup_{\|x\|_X \leq 1} |\varphi(x)|.$$

Remark 5.76: Existence and Uniqueness of the Dual Space. For every normed vector space X , the dual space X^* exists in the set-theoretic sense: it is simply the collection of all bounded linear maps from X into the scalar field $\mathbb{F}$. At least one such functional always exists, namely the zero functional

$$0(x) = 0.$$

The deeper question is not whether X^* exists, but whether it contains enough nonzero functionals to detect the geometry of X . This is one of the reasons **the Hahn–Banach theorem** is important: it guarantees many continuous linear functionals on normed spaces.

The dual space is also unique once X and the scalar field are fixed:

$$X^* = L(X, \mathbb{F})$$

is a definition, not a construction with choices. If two authors build the dual space using the same bounded linear functionals and the same operator norm, they have the same normed space. If they present it in different notation, the identification is canonical.

5.9.1 The Banach Space of Bounded Linear Operators

Intuition / First Principles. An operator-norm Cauchy sequence (A_n) means the operators eventually act almost the same on every unit vector. Therefore, for each fixed $x \in X$, the outputs $A_n x$ should converge in the target space Y . The theorem works because Y is complete, so those pointwise output limits actually exist.

First-principles thinking. Notice where completeness is used: not in the domain X , but in the target Y . We are not taking limits of vectors inside X ; we are taking limits of the output sequence $(A_n x)$ inside Y .

Remark 5.77: Source. The next theorem is Theorem 1.3.1 in Theo Bühler and Dietmar A. Salamon, *Functional Analysis*. The material below adds the local definitions and no-skipped-steps estimates needed to read their proof comfortably.

Definition 5.78: Bounded Operators and the Operator Norm, Local Reminder.

Let X and Y be normed vector spaces over the same field. A linear map $T : X \rightarrow Y$ is *bounded* if there exists a constant $C \geq 0$ such that

$$\|Tx\|_Y \leq C \|x\|_X \quad \text{for every } x \in X.$$

The space of bounded linear operators from X to Y is

$$L(X, Y) = \{T : X \rightarrow Y : T \text{ is linear and bounded}\}.$$

For $T \in L(X, Y)$, the *operator norm* is

$$\|T\|_{L(X, Y)} = \sup_{\|x\|_X \leq 1} \|Tx\|_Y.$$

It measures the largest output size that T can produce from inputs of size at most 1.

Lemma 5.79: Basic Operator-Norm Estimate.

If $T \in L(X, Y)$, then

$$\|Tx\|_Y \leq \|T\|_{L(X, Y)} \|x\|_X \quad \text{for every } x \in X.$$

Proof. If $x = 0$, then $Tx = T0 = 0$, so the estimate is immediate. Now suppose $x \neq 0$ and define

$$u = \frac{x}{\|x\|_X}.$$

Then $\|u\|_X = 1$. Since u is in the unit ball, the definition of operator norm gives

$$\|Tu\|_Y \leq \|T\|_{L(X, Y)}. \quad ([\text{unit ball}] \ 1)$$

Because $x = \|x\|_X u$, linearity gives

$$\begin{aligned} \|Tx\|_Y &= \|T(\|x\|_X u)\|_Y && ([\text{rewrite } x] \ 2) \\ &= \|\|x\|_X Tu\|_Y && ([\text{linearity}] \ 3) \\ &= \|x\|_X \|Tu\|_Y && ([\text{homogeneity}] \ 4) \\ &\leq \|x\|_X \|T\|_{L(X, Y)}. && ([\text{use (1)}] \ 5) \end{aligned}$$

This proves the estimate. $\square$

Definition 5.80: Cauchy in Operator Norm.

A sequence (A_n) in $L(X, Y)$ is Cauchy in operator norm if

$$\forall \rho > 0, \exists N \in \mathbb{N}, \forall m, n \geq N, \quad \|A_m - A_n\|_{L(X, Y)} < \rho.$$

The tolerance ρ lives in the operator space $L(X, Y)$. When we fix one input x , this operator error will be converted into an output error inside Y .

Theorem 5.81: Completeness of Operator Spaces.

Let X be a normed vector space and let Y be a Banach space over the same field. Then $L(X, Y)$ is a Banach space with respect to the operator norm.

Proof. Let $(A_n)_{n \in \mathbb{N}}$ be a Cauchy sequence in $L(X, Y)$. We must show that it converges, in operator norm, to some bounded linear operator $A : X \rightarrow Y$.

Step 1: operator Cauchy implies pointwise Cauchy. Fix $x \in X$. We prove that $(A_n x)$ is Cauchy in Y . Let $\eta > 0$ be arbitrary. Choose

$$\rho = \frac{\eta}{\|x\|_X + 1}. \quad ([\text{choose operator tolerance}] 1)$$

This number is positive because $\eta > 0$ and $\|x\|_X + 1 > 0$. Since (A_n) is Cauchy in operator norm, there exists $N \in \mathbb{N}$ such that

$$m, n \geq N \implies \|A_m - A_n\|_{L(X, Y)} < \frac{\eta}{\|x\|_X + 1}. \quad ([\text{Cauchy choice of } N] 2)$$

Then, for $m, n \geq N$,

$$\begin{aligned} \|A_m x - A_n x\|_Y &= \|(A_m - A_n)x\|_Y && ([\text{linearity of difference}] 3) \\ &\leq \|A_m - A_n\|_{L(X, Y)} \|x\|_X && ([\text{Lemma 5.79}] 4) \\ &< \frac{\eta}{\|x\|_X + 1} \|x\|_X && ([\text{use (2)}] 5) \\ &= \eta \frac{\|x\|_X}{\|x\|_X + 1} && ([\text{algebra}] 6) \\ &< \eta. && ([t/(t+1) < 1] 7) \end{aligned}$$

Therefore $(A_n x)$ is Cauchy in Y for every fixed $x \in X$.

Step 2: use completeness of Y to define the candidate limit. Because Y is Banach, every Cauchy sequence in Y converges in Y . Hence, for every fixed $x \in X$, the sequence $(A_n x)$ has a limit in Y . Define

$$Ax := \lim_{n \rightarrow \infty} A_n x. \quad (\text{BS 1.3.1})$$

This gives a map $A : X \rightarrow Y$. This is the only place where the Banach assumption on Y is essential: the limits are taken in the output space Y , not in the domain X .

Step 3: prove that the pointwise limit is linear. Let $x, z \in X$ and let α, β be scalars. Then

$$\begin{aligned} A(\alpha x + \beta z) &= \lim_{n \rightarrow \infty} A_n(\alpha x + \beta z) && ([\text{definition of } A] 3) \\ &= \lim_{n \rightarrow \infty} (\alpha A_n x + \beta A_n z) && ([\text{linearity of } A_n] 4) \\ &= \alpha \lim_{n \rightarrow \infty} A_n x + \beta \lim_{n \rightarrow \infty} A_n z && ([\text{limit algebra}] 5) \\ &= \alpha Ax + \beta Az. && ([\text{definition of } A] 6) \end{aligned}$$

Thus A is linear.

Step 4: prove that A is bounded. To prove $A \in L(X, Y)$, we need one constant C such that

$$\|Ax\|_Y \leq C \|x\|_X \quad \text{for every } x \in X.$$

Apply the operator-norm Cauchy condition with tolerance 1. There exists $N_0 \in \mathbb{N}$ such that

$$m \geq N_0 \implies \|A_m - A_{N_0}\|_{L(X, Y)} < 1. \quad ([\text{Cauchy with tolerance 1}] \ 8)$$

Fix $x \in X$. For every $m \geq N_0$,

$$\|A_mx - A_{N_0}x\|_Y = \|(A_m - A_{N_0})x\|_Y \quad ([\text{difference}] \ 9)$$

$$\leq \|A_m - A_{N_0}\|_{L(X, Y)} \|x\|_X \quad ([\text{Lemma 5.79}] \ 10)$$

$$< \|x\|_X. \quad ([\text{use (8)}] \ 11)$$

Let $m \rightarrow \infty$. Since $A_mx \rightarrow Ax$ in Y , we get

$$A_mx - A_{N_0}x \rightarrow Ax - A_{N_0}x. \quad ([\text{subtract fixed vector}] \ 12)$$

The norm is continuous, so

$$\|Ax - A_{N_0}x\|_Y \leq \|x\|_X. \quad ([\text{limit of (11)}] \ 13)$$

Now use the triangle inequality:

$$\|Ax\|_Y = \|(Ax - A_{N_0}x) + A_{N_0}x\|_Y \quad ([\text{add and subtract}] \ 14)$$

$$\leq \|Ax - A_{N_0}x\|_Y + \|A_{N_0}x\|_Y \quad ([\text{triangle}] \ 15)$$

$$\leq \|x\|_X + \|A_{N_0}\|_{L(X, Y)} \|x\|_X \quad ([\text{use (13) and boundedness}] \ 16)$$

$$= (1 + \|A_{N_0}\|_{L(X, Y)}) \|x\|_X. \quad ([\text{factor}] \ 17)$$

So A is bounded. Since Step 3 already proved that A is linear, we have $A \in L(X, Y)$.

Step 5: prove convergence in operator norm. Let $\varepsilon > 0$. Since (A_n) is Cauchy in operator norm, there exists $N \in \mathbb{N}$ such that

$$m, n \geq N \implies \|A_m - A_n\|_{L(X, Y)} < \varepsilon. \quad ([\text{Cauchy choice of } N] \ 18)$$

Fix $n \geq N$ and $x \in X$. For every $m \geq N$,

$$\|A_mx - A_nx\|_Y = \|(A_m - A_n)x\|_Y \quad ([\text{difference}] \ 19)$$

$$\leq \|A_m - A_n\|_{L(X, Y)} \|x\|_X \quad ([\text{Lemma 5.79}] \ 20)$$

$$< \varepsilon \|x\|_X. \quad ([\text{use (18)}] \ 21)$$

Let $m \rightarrow \infty$. Since $A_mx \rightarrow Ax$ in Y and the norm is continuous,

$$\|(A - A_n)x\|_Y = \|Ax - A_nx\|_Y \leq \varepsilon \|x\|_X \quad (x \in X). \quad (\text{BS 1.3.2})$$

Taking the supremum over all $\|x\|_X \leq 1$,

$$\|A - A_n\|_{L(X, Y)} = \sup_{\|x\|_X \leq 1} \|(A - A_n)x\|_Y \quad ([\text{op norm}] \ 22)$$

$$\leq \sup_{\|x\|_X \leq 1} \varepsilon \|x\|_X \quad ([\text{use (BS 1.3.2)}] \ 23)$$

$$\leq \varepsilon. \quad ([\text{unit ball}] \ 24)$$

Therefore, for each $\varepsilon > 0$, there is $N \in \mathbb{N}$ such that $\|A - A_n\|_{L(X,Y)} \leq \varepsilon$ whenever $n \geq N$. Hence $A_n \rightarrow A$ in operator norm. Every Cauchy sequence in $L(X, Y)$ therefore converges to an element of $L(X, Y)$, so $L(X, Y)$ is Banach. $\square$

Corollary 5.82: Dual Spaces Are Banach.

If X is a normed vector space over $\mathbb{F}$, where $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$, then its dual space $X^* = L(X, \mathbb{F})$ is a Banach space.

Proof. Apply Theorem 5.81 with $Y = \mathbb{F}$. The scalar field $\mathbb{R}$ or $\mathbb{C}$ is complete, so the target is Banach. Therefore $L(X, \mathbb{F}) = X^*$ is Banach. $\square$

Remark 5.83: Why This Matters. The original space X does not have to be complete for its dual space to be complete. Completeness of X^* comes from the completeness of the scalar target $\mathbb{F}$. This is why dual spaces are stable objects: even when X has missing limits, its bounded linear measurements still form a Banach space.

Takeaway. The dual space is not just a list of functionals. It is a Banach space in its own right, with distance measured by operator norm:

$$\|\varphi - \psi\| = \sup_{\|x\|_X \leq 1} |\varphi(x) - \psi(x)|.$$

5.10 Integral Operators and Green's Functions

Big Idea. Integral functionals collapse a function to one number. Integral operators keep a free variable alive, so the output is a whole new function.

Remark 5.84: Fredholm and Volterra Kernels. Fredholm integral equations and Volterra integral equations usually belong first to the world of *integral operators*, not merely integral functionals.

A Fredholm-type kernel $K(x, t)$ defines an operator

$$(Tf)(x) = \int_a^b K(x, t)f(t) dt.$$

Here the input is a function f , and the output is again a function of x . Thus

$$T : X \rightarrow Y$$

is an operator between function spaces, so T is a point of an operator space such as $L(X, Y)$ when it is bounded and linear.

A Volterra-type kernel has a variable upper limit:

$$(Vf)(x) = \int_a^x K(x, t)f(t) dt.$$

This is still an operator $V : f \mapsto Vf$, because the output depends on x .

Intuition / First Principles. The quick test is: look at the output. If the rule sends a function to a number, it is a functional. If the rule sends a function to another function, it is an operator.

First-principles thinking. A functional collapses all the information in f into one scalar:

$$f \mapsto \int_0^1 f(t)g(t) dt.$$

An integral operator keeps a free variable x alive:

$$f \mapsto \left(x \mapsto \int_a^b K(x, t)f(t) dt \right).$$

So a kernel $K(x, t)$ is like a whole family of functionals, one for each fixed x .

Example 5.85: Fixed x Gives a Functional. For each fixed x , the rule

$$f \mapsto \int_a^b K(x, t)f(t) dt$$

is a scalar-valued functional of f . But collecting all those scalars as x varies gives the function

$$x \mapsto (Tf)(x).$$

That collection is the integral operator T . This is why Fredholm and Volterra kernels are

naturally studied in operator theory, while the single integral $I_g(f)$ above is a functional. ■

Example 5.86: Green's Function for an ODE BVP. Consider the boundary value problem

$$-u''(x) = f(x), \quad u(0) = u(1) = 0.$$

The Green's function $G(x, t)$ is defined by

$$-\frac{\partial^2}{\partial x^2} G(x, t) = \delta_t(x), \quad G(0, t) = G(1, t) = 0.$$

For this problem,

$$G(x, t) = \begin{cases} x(1-t), & 0 \leq x \leq t, \\ t(1-x), & t \leq x \leq 1. \end{cases}$$

Then the solution is

$$u(x) = \int_0^1 G(x, t) f(t) dt.$$

For each fixed x , the rule

$$f \mapsto u(x) = \int_0^1 G(x, t) f(t) dt$$

is a functional of f : it produces one number, the value of the solution at that particular x . The full Green's formula keeps x free:

$$T : f \mapsto u = Tf, \quad (Tf)(x) = u(x) = \int_0^1 G(x, t) f(t) dt, \quad 0 \leq x \leq 1.$$

For the explicit Green's function above, this is

$$u(x) = (1-x) \int_0^x t f(t) dt + x \int_x^1 (1-t) f(t) dt.$$

This is the integral operator: it sends the input function f to the whole output function u . ■

Intuition / First Principles. Green's functions are the bridge between point sources and operators. The equation

$$L_x G(x, t) = \delta_t(x)$$

says: fix the source point t , hit the system with a unit point source there, and record the response as a function of x .

First-principles thinking. A general forcing $f(t)$ is built by adding many tiny point sources. The response to a point source at t is $G(x, t)$, so the total response is the superposition

$$u(x) = \int G(x, t) f(t) dt.$$

Thus the delta appears in the definition of G , while the final solution formula is an integral operator.

Takeaway. Green's formulas belong in operator theory because the output is the entire solution function u , not merely one scalar value of u .

5.11 Hilbert Spaces and Riesz Representation

Big Idea. Hilbert spaces add inner-product geometry to complete normed spaces. That extra geometry lets every continuous linear functional be represented by inner product with a unique vector.

Theorem 5.87: Riesz Representation for Hilbert Spaces.

Let H be a Hilbert space. For every continuous linear functional $F \in H^*$, there exists a unique vector $g \in H$ such that

$$F(f) = \langle f, g \rangle \quad \text{for every } f \in H.$$

Remark 5.88. Riesz does not say that every function has a unique functional in every possible function space. It says that in a Hilbert space, every continuous linear functional can be represented uniquely by an inner product with some vector. In $L^2(\Omega)$, this means every $F \in (L^2(\Omega))^*$ has the form

$$F(f) = \int_{\Omega} f(x) \overline{g(x)} dx$$

for a unique $g \in L^2(\Omega)$. So g is a function-as-point, and F is a functional-as-point; Riesz gives a one-to-one correspondence between the two roles.

Remark 5.89: Inner Products and Functionals in Quantum Mechanics. The quantum-mechanical intuition is correct, but the precise statement is this: an inner product is not itself a functional on H , because it takes two inputs,

$$\langle \cdot, \cdot \rangle : H \times H \rightarrow \mathbb{C}.$$

It is a scalar-valued form. But if we fix one vector, it becomes a functional of the other vector. In Dirac notation, a ket $|\psi\rangle$ is a vector in the Hilbert space H . The corresponding bra $\langle\psi|$ is the functional

$$\langle\psi| : H \rightarrow \mathbb{C}, \quad \langle\psi|(|\phi\rangle) = \langle\psi, \phi\rangle.$$

So a bra is literally a linear functional on the state space. The number

$$\langle\psi, \phi\rangle$$

is the value of that functional applied to $|\phi\rangle$.

Intuition / First Principles. QM turns the Riesz representation theorem into notation. A state can be viewed as a vector $|\psi\rangle$, but the same state can also be used to test another state by inner product:

$$|\phi\rangle \mapsto \langle\psi, \phi\rangle.$$

First-principles thinking. A vector contains geometric information. A functional asks a scalar question about an input vector. The inner product is the machine that turns a fixed vector into

such a question. In finite dimensions, this is the familiar move from a column vector to a row vector; in Hilbert spaces, Riesz says the same idea still works.

Takeaway. Use *function space* for spaces like $C([0, 1])$ and $L^2(\Omega)$, whose points are functions. Use *dual space* or *space of functionals* for spaces like X^* , whose points are scalar-valued maps on X . In Hilbert spaces, Riesz explains why these two worlds can be identified, but they should still be conceptually distinguished.

CHAPTER 6

Probability Theory

First Topics

1. Probability spaces as measure spaces.
2. Random variables as measurable maps.
3. Expectation, conditional expectation, and convergence.
4. Stochastic processes as $X(t, \omega)$.
5. Filtrations and adapted processes.
6. Brownian motion.
7. Martingales and stopping times.
8. Stochastic integrals.
9. Ito's formula.
10. Stochastic differential equations and diffusions.
11. Heston and rough Heston stochastic volatility models.
12. Girsanov's theorem and Feynman–Kac formula.

Learning Objectives.

- See probability theory as measure theory with total mass one.
- Treat random variables as measurable maps out of a probability space.
- Understand a stochastic process as a family of random variables, or equivalently as a two-variable map $X(t, \omega)$.
- Understand Brownian motion as the basic continuous random path: continuous everywhere, but differentiable nowhere with probability one.
- Learn why ordinary calculus fails for Brownian paths and why stochastic integration must be built using quadratic-mean ideas.
- Study stochastic processes, martingales, filtrations, and stopping times as the bookkeeping for evolving random systems.
- Understand stochastic differential equations and diffusions as random analogues of ordinary differential equations.
- Use Ito's formula as the chain rule for stochastic calculus.
- Understand stochastic volatility models, especially the Heston model and the rough Heston model.
- Learn how changes of probability measure enter through Girsanov's theorem.
- Connect stochastic processes with PDE through the Feynman–Kac formula.

Core Questions

- What does it mean for a system to evolve randomly over time?
 - Why does Brownian motion require a new kind of calculus?
 - How do martingales encode fair-game structure and conditional information?
 - Why do financial models make volatility random?
 - What changes when volatility is rough rather than Markovian?
 - How can expectations of random paths solve deterministic PDEs?
-

Foundations to Strengthen.

- Start from probability spaces as measure spaces with total mass one.
- Read random variables as measurable maps and laws as pushforward measures.
- Learn conditional expectation before martingales.
- Separate almost sure convergence, convergence in probability, and L^p convergence.
- Before stochastic calculus, know filtrations, adapted processes, stopping times, and martingales.

Functional-Analysis Lens for This Chapter. Probability becomes easier to organize if random variables are treated as points in function spaces. The basic structural translation is:

$$\text{random variable} \longleftrightarrow \text{measurable function on } (\Omega, \mathcal{F}, \mathbb{P}).$$

Then $L^p(\Omega)$ is the space of random variables whose p th moment is finite:

$$\|X\|_{L^p} = (\mathbb{E}|X|^p)^{1/p}.$$

So L^p convergence is norm convergence of random variables. In particular, $L^2(\Omega)$ is a Hilbert space with inner product

$$\langle X, Y \rangle_{L^2} = \mathbb{E}[XY].$$

This viewpoint explains why functional analysis keeps appearing: conditional expectation is a projection, martingales are compatible families of projections, stochastic integration is built using L^2 limits, and Markov processes lead to semigroups and generators.

Probability object	Functional-analysis reading
Random variable X	point of a function space such as $L^p(\Omega)$
Expectation $\mathbb{E}[X]$	integral, hence a linear functional when defined
Variance and covariance	norm and inner product after centering
Conditional expectation	projection onto a smaller information subspace
Martingale	process consistent with those projections over time
Ito integral	limit in a norm, usually built first in L^2
Markov process	transition operators, semigroups, and generators

6.1 Probability Space as a Measure Space

Big Idea. Probability theory begins as measure theory with a special normalization: the whole universe has measure one. The symbol changes from μ to $\mathbb{P}$, but the underlying structure is the same.

Definition 6.1: Probability Space.

A probability space is a triple

$$(\Omega, \mathcal{F}, \mathbb{P}),$$

where

- Ω is the sample space, the set of all possible outcomes;
- $\mathcal{F}$ is a σ -algebra of events, the subsets of Ω whose probabilities are allowed to be measured;
- $\mathbb{P} : \mathcal{F} \rightarrow [0, 1]$ is a measure with

$$\mathbb{P}(\Omega) = 1.$$

Thus a probability space is exactly a measure space whose total mass is one.

Remark 6.2: Measure-Theory Lens. This definition is not new machinery beyond measure theory. It is the same object

$$(\Omega, \mathcal{F}, \mu),$$

with the extra normalization $\mu(\Omega) = 1$. The word “probability” adds interpretation: measurable sets are events, and the measure of an event is its probability.

Remark 6.3: Why the Sigma-Algebra Is There. In elementary probability, one often writes $\mathbb{P}(A)$ for any subset $A \subseteq \Omega$. Measure theory teaches us to be more careful: $\mathbb{P}(A)$ is defined only when $A \in \mathcal{F}$. The σ -algebra is the list of questions about the outcome that the model can answer.

Example 6.4: Coin Toss. For one coin toss,

$$\Omega = \{H, T\}, \quad \mathcal{F} = \{\emptyset, \{H\}, \{T\}, \Omega\},$$

and, for a fair coin,

$$\mathbb{P}(\{H\}) = \mathbb{P}(\{T\}) = \frac{1}{2}.$$

This is a finite measure space with total mass one. ■

6.2 Random Variables as Measurable Maps

Big Idea. A random variable is not random because it is mysterious. It is a function on the outcome space. Randomness comes from not knowing which $\omega \in \Omega$ occurred.

Definition 6.5: Random Variable.

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. A real-valued random variable is a measurable map

$$X : \Omega \rightarrow \mathbb{R}.$$

Measurable means that for every Borel set $B \subseteq \mathbb{R}$,

$$X^{-1}(B) = \{\omega \in \Omega : X(\omega) \in B\} \in \mathcal{F}.$$

So every question about the value of X must correspond to an event in Ω .

Remark 6.6: Measure-Theory Lens. A random variable is a measurable function. The condition $X^{-1}(B) \in \mathcal{F}$ says that every observable question about the output value of X must be answerable inside the original probability space. Probability enters only after we measure those preimages:

$$\mathbb{P}(X \in B) = \mathbb{P}(X^{-1}(B)).$$

Remark 6.7: Measurable Versus Integrable. Measurable and integrable are different levels of structure.

Measurable means that X is a legitimate random variable: for every reasonable set of output values $B \subseteq \mathbb{R}$, the question “did X land in B ?” corresponds to an event

$$\{\omega : X(\omega) \in B\} \in \mathcal{F}.$$

So measurability is about whether probabilities of output events are defined.

Integrable is stronger. It means that the average size is finite:

$$\mathbb{E}|X| = \int_{\Omega} |X(\omega)| d\mathbb{P}(\omega) < \infty.$$

Only then does $\mathbb{E}[X]$ exist as a finite number. A random variable can be measurable but not integrable.

Remark 6.8: Why $X : \Omega \rightarrow \mathbb{R}$, Not $\mathcal{F} \rightarrow \mathbb{R}$? The sample point $\omega \in \Omega$ is the actual outcome. A random variable assigns a numerical value to each possible outcome:

$$\omega \mapsto X(\omega).$$

The σ -algebra $\mathcal{F}$ is not the set of outcomes. It is the collection of events, meaning subsets of Ω whose probabilities are allowed to be measured. Thus $\mathcal{F}$ is the bookkeeping system for questions, while Ω is the space on which the function actually lives.

For example, in one coin toss, the outcome is H or T . A payoff random variable might be

$$X(H) = 1, \quad X(T) = -1.$$

That is a function on outcomes. The event $\{H\} \in \mathcal{F}$ is then used to ask and measure questions such as “is $X = 1$?”

Remark 6.9: Distribution. The distribution, or law, of X is the pushforward measure

$$\mathcal{L}_X(B) = \mathbb{P}(X \in B) = \mathbb{P}(X^{-1}(B)).$$

This moves probability from the abstract sample space Ω onto the more familiar real line.

Definition 6.10: Expectation.

If X is integrable, its expectation is the integral of X over the probability space:

$$\mathbb{E}[X] = \int_{\Omega} X(\omega) d\mathbb{P}(\omega).$$

Thus expectation is not a new operation separate from analysis. It is Lebesgue integration with respect to a probability measure.

Remark 6.11: Analysis Lens. Expectation is a linear operation on integrable random variables:

$$\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y].$$

So, on a space such as $L^1(\Omega)$, expectation is a linear functional. The probabilistic word is “average,” but the analytic object is an integral.

Remark 6.12: Expectation Versus Inner Product. Expectation by itself,

$$X \mapsto \mathbb{E}[X],$$

is a linear functional, not an inner product. The inner product appears when two square-integrable random variables are paired:

$$\langle X, Y \rangle_{L^2} = \mathbb{E}[XY] \quad \text{for real-valued random variables.}$$

For complex-valued random variables, use

$$\langle X, Y \rangle_{L^2} = \mathbb{E}[X\overline{Y}].$$

Strictly speaking, $L^2(\Omega)$ identifies random variables that are equal almost surely. With that identification, this pairing is an inner product. After centering, covariance is exactly this inner product:

$$\text{Cov}(X, Y) = \langle X - \mathbb{E}X, Y - \mathbb{E}Y \rangle_{L^2}.$$

Takeaway. The dictionary is:

measure theory	probability theory
measure space $(\Omega, \mathcal{F}, \mu)$	probability space $(\Omega, \mathcal{F}, \mathbb{P})$
measurable set	event
measurable function	random variable
integral	expectation

6.3 Stochastic Processes as Families of Random Variables

Big Idea. A stochastic process is a random object evolving in time. Formally, it is a family of random variables indexed by time:

$$(X_t)_{t \in T}.$$

Equivalently, it is a two-variable map

$$X : T \times \Omega \rightarrow \mathbb{R}, \quad (t, \omega) \mapsto X(t, \omega).$$

The two variables have different meanings: t is time, and ω is the outcome of the random experiment.

Definition 6.13: Stochastic Process.

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let T be an index set, usually $T = \mathbb{N}$, $T = \mathbb{Z}$, or $T = [0, \infty)$. A stochastic process is a family

$$(X_t)_{t \in T}$$

such that each

$$X_t : \Omega \rightarrow \mathbb{R}$$

is a random variable. Equivalently, a process is a map

$$X : T \times \Omega \rightarrow \mathbb{R}, \quad X_t(\omega) = X(t, \omega).$$

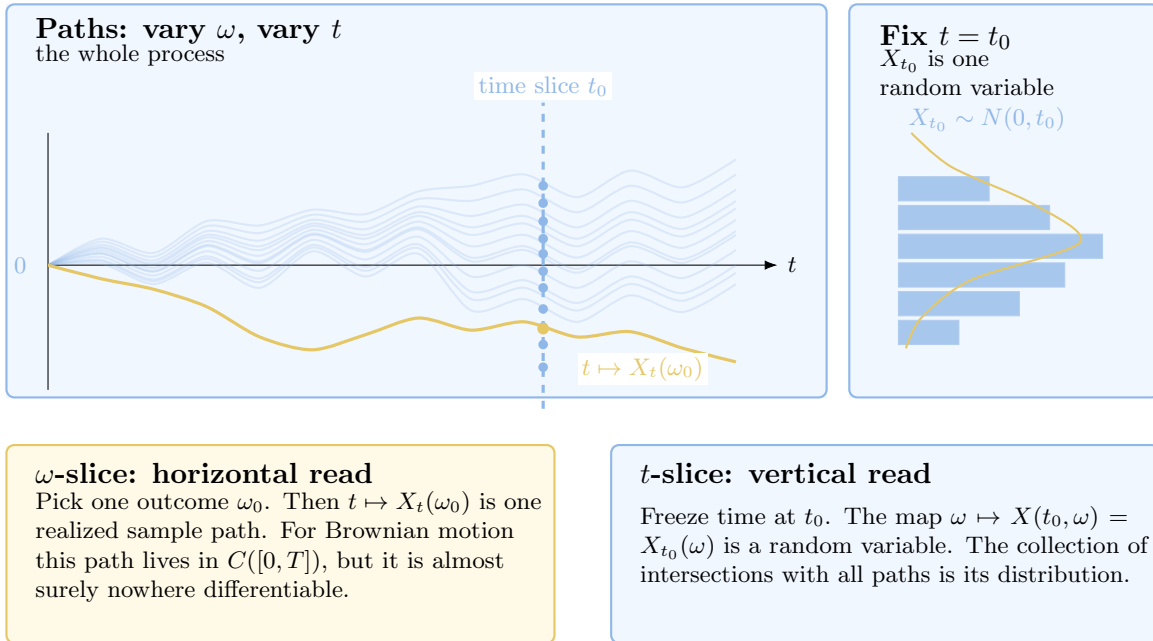
Remark 6.14: Function-Space Lens. For each fixed t , X_t is a point of a space such as $L^p(\Omega)$. For each fixed ω , the map $t \mapsto X(t, \omega)$ is a path. Thus a stochastic process can be read as a curve in a function space:

$$t \mapsto X_t \in L^p(\Omega),$$

or as a random element of a path space such as $C([0, T])$. Much of stochastic analysis is learning which of these readings is being used.

Reading a stochastic process $X(t, \omega)$

One process can be read in two directions: along time for one outcome, or vertically at one time across many outcomes.



Takeaway. The notation X_t hides two readings. Fixing t gives a random variable on Ω ; fixing ω gives one path through time. Stochastic calculus studies both at once.

Remark 6.15: The Two Slices. The picture has two different slices.

Fix time $t = t_0$: $\omega \mapsto X(t_0, \omega)$ is a random variable.

Fix outcome $\omega = \omega_0$: $t \mapsto X(t, \omega_0)$ is one realized path.

This is the key mental model. A process is not one function of time; it is a random family of functions of time.

Example 6.16: Random Walk. Let $\xi_1, \xi_2, \dots$ be independent random variables with

$$\mathbb{P}(\xi_k = 1) = \mathbb{P}(\xi_k = -1) = \frac{1}{2}.$$

Define

$$S_n = \sum_{k=1}^n \xi_k.$$

Then $(S_n)_{n \geq 0}$ is a discrete-time stochastic process. For each fixed n , S_n is a random variable. For each fixed outcome ω , the sequence

$$n \mapsto S_n(\omega)$$

is one path of the random walk. ■

6.4 Filtrations: Information Growing in Time

Big Idea. Stochastic calculus is not only about random paths. It is also about what information is available at each time. A filtration records this information.

Definition 6.17: Filtration.

A filtration is an increasing family of σ -algebras

$$(\mathcal{F}_t)_{t \geq 0}$$

such that

$$s \leq t \implies \mathcal{F}_s \subseteq \mathcal{F}_t.$$

The interpretation is that $\mathcal{F}_t$ contains all events whose truth can be determined using information available up to time t .

Remark 6.18: Measure-Theory Lens. A filtration is not a sequence of numbers; it is a sequence of σ -algebras. Since larger σ -algebras contain more measurable sets, the inclusion

$$\mathcal{F}_s \subseteq \mathcal{F}_t$$

means that later times allow more events to be observed. Information is modeled by measurability.

Definition 6.19: Adapted Process.

A process $(X_t)_{t \geq 0}$ is adapted to a filtration $(\mathcal{F}_t)_{t \geq 0}$ if

$$X_t \text{ is } \mathcal{F}_t\text{-measurable for every } t.$$

This means that the value X_t is known at time t . An adapted process does not look into the future.

Remark 6.20: Measure-Theory Lens. Adaptedness is a measurability condition. Saying that X_t is $\mathcal{F}_t$ -measurable means every event of the form

$$\{X_t \in B\}$$

belongs to the information available by time t . This is why “no future information” becomes a precise mathematical condition.

Remark 6.21: Why This Matters. In ordinary calculus, an integral can use the whole function at once. In stochastic calculus, the integrand must usually be adapted: at time t , the decision about how much to multiply by dB_t can use the past and present, but not future Brownian motion. This is the mathematical version of “no future information.”

6.5 Brownian Motion

Big Idea. Brownian motion is the canonical continuous-time random path. It is the limit object behind random walks and the noise source in many stochastic differential equations.

Definition 6.22: Brownian Motion.

A process $(B_t)_{t \geq 0}$ is a standard Brownian motion if

- $B_0 = 0$;
- $B_t - B_s \sim N(0, t - s)$ whenever $0 \leq s < t$;
- increments over disjoint time intervals are independent;
- the paths $t \mapsto B_t(\omega)$ are continuous with probability one.

Remark 6.23: Analysis Lens. Brownian motion is simultaneously an analytic and probabilistic object. Analytically, its sample paths live in $C([0, T])$ with probability one. Probabilistically, each fixed B_t is a Gaussian random variable. The central tension is that the paths are continuous but much rougher than ordinary differentiable curves.

Remark 6.24: The Strange Part. Brownian paths are continuous, but with probability one they are nowhere differentiable. So the expression

$$\frac{dB_t}{dt}$$

does not exist as an ordinary derivative. This is the obstruction that forces stochastic calculus to differ from ordinary calculus.

Definition 6.25: Quadratic Variation.

For a partition

$$0 = t_0 < t_1 < \cdots < t_n = T,$$

the quadratic variation sum of a path is

$$\sum_{k=0}^{n-1} (B_{t_{k+1}} - B_{t_k})^2.$$

For Brownian motion, as the mesh of the partition goes to zero,

$$\sum_{k=0}^{n-1} (B_{t_{k+1}} - B_{t_k})^2 \longrightarrow T$$

in probability, and in stronger senses under appropriate refinements.

Remark 6.26: Analysis Lens. Quadratic variation measures roughness at second order. Smooth paths have zero quadratic variation; Brownian paths have nonzero quadratic variation. This is the analytic reason stochastic calculus differs from ordinary calculus: small Brownian increments are too large, in aggregate, for their squares to disappear.

Remark 6.27: Why Ordinary Calculus Fails. For a smooth path $x(t)$, the quadratic variation is zero:

$$\sum (x(t_{k+1}) - x(t_k))^2 \rightarrow 0.$$

For Brownian motion, the quadratic variation is nonzero:

$$[B]_T = T.$$

This single fact changes the chain rule. In stochastic calculus, second-derivative terms survive.

6.6 Ito Integration

Big Idea. The stochastic integral

$$\int_0^T H_t dB_t$$

is not defined by ordinary Riemann integration, because Brownian motion is too rough. It is built first for simple adapted processes, then extended by an L^2 limit.

Definition 6.28: Ito Integral for Step Processes.

Let H_t be an adapted step process of the form

$$H_t = \sum_{k=0}^{n-1} H_k 1_{(t_k, t_{k+1}]}(t),$$

where each H_k is $\mathcal{F}_{t_k}$ -measurable. Define

$$\int_0^T H_t dB_t = \sum_{k=0}^{n-1} H_k (B_{t_{k+1}} - B_{t_k}).$$

The coefficient H_k is chosen using information available at the left endpoint t_k . This left-endpoint rule is what makes the integral non-anticipating.

Remark 6.29: Measure-Theory and Functional-Analysis Lens. The left endpoint condition $H_k \in \mathcal{F}_{t_k}$ is the measure theory part: the integrand uses only information already available. The extension from step processes to general integrands is the functional analysis part: define the integral on a dense class, prove a norm estimate, then complete the space.

Theorem 6.30: Ito Isometry.

For square-integrable adapted processes,

$$\mathbb{E} \left[\left(\int_0^T H_t dB_t \right)^2 \right] = \mathbb{E} \left[\int_0^T H_t^2 dt \right].$$

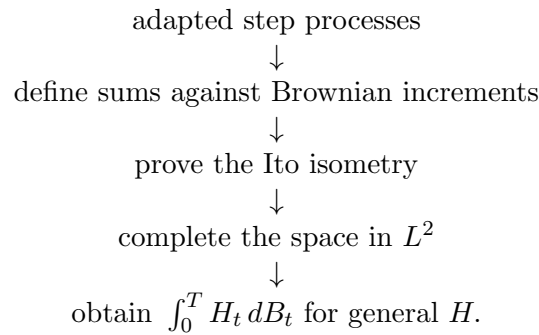
This identity is the basic estimate that allows the Ito integral to be constructed by completion in L^2 .

Remark 6.31: Functional-Analysis Lens. This theorem says the Ito integral is controlled by an L^2 norm:

$$\left\| \int_0^T H_t dB_t \right\|_{L^2(\Omega)}^2 = \mathbb{E} \int_0^T H_t^2 dt.$$

So the stochastic integral is first an isometry from a space of integrands into $L^2(\Omega)$. That is why the construction is a Hilbert-space construction, not a pathwise Riemann integral.

Remark 6.32: The Construction in Words. The construction follows this path:



This is why Hilbert-space thinking is hidden inside stochastic calculus: the integral is built using an L^2 norm and an isometry.

6.7 Ito Formula and Stochastic Differential Equations

Big Idea. Ito's formula is the stochastic chain rule. It has an extra second-derivative term because Brownian motion has nonzero quadratic variation.

Theorem 6.33: Ito Formula, One-Dimensional Form.

Suppose

$$dX_t = b(t, X_t) dt + \sigma(t, X_t) dB_t.$$

If $f(t, x)$ is sufficiently smooth, then

$$\begin{aligned} df(t, X_t) &= f_t(t, X_t) dt + f_x(t, X_t) dX_t + \frac{1}{2} f_{xx}(t, X_t) \sigma(t, X_t)^2 dt \\ &= \left(f_t + b f_x + \frac{1}{2} \sigma^2 f_{xx} \right) (t, X_t) dt + \sigma(t, X_t) f_x(t, X_t) dB_t. \end{aligned}$$

Remark 6.34: Analysis Lens. Ito's formula is Taylor expansion plus quadratic variation. The first derivative terms behave like ordinary calculus, but the second derivative term survives because

$$(dB_t)^2 = dt.$$

So the theorem is a chain rule adapted to paths whose first-order variation is too wild but whose quadratic variation is controlled.

Remark 6.35: Where the Extra Term Comes From. In ordinary calculus, one thinks of $(dt)^2$ as negligible. In Ito calculus, the mnemonic rules are

$$(dt)^2 = 0, \quad dt dB_t = 0, \quad (dB_t)^2 = dt.$$

The last identity is the informal expression of Brownian quadratic variation.

Definition 6.36: Stochastic Differential Equation.

A stochastic differential equation, or SDE, has the form

$$dX_t = b(t, X_t) dt + \sigma(t, X_t) dB_t.$$

Equivalently, it means the integral equation

$$X_t = X_0 + \int_0^t b(s, X_s) ds + \int_0^t \sigma(s, X_s) dB_s.$$

The first integral is an ordinary time integral. The second is an Ito integral.

Remark 6.37: Analysis Lens. An SDE should be read through its integral form. The differential notation is compressed notation for two maps:

$$X \mapsto \int_0^t b(s, X_s) ds, \quad X \mapsto \int_0^t \sigma(s, X_s) dB_s.$$

Existence and uniqueness questions are therefore fixed-point questions in spaces of adapted processes.

Example 6.38: Geometric Brownian Motion. The model

$$dS_t = \mu S_t dt + \sigma S_t dB_t$$

has solution

$$S_t = S_0 \exp \left(\left(\mu - \frac{1}{2} \sigma^2 \right) t + \sigma B_t \right).$$

The correction $-\frac{1}{2}\sigma^2 t$ is exactly an Ito correction. It is not present in the ordinary chain rule. ■

6.8 Stochastic Volatility: Heston and Rough Heston

Big Idea. Geometric Brownian motion assumes that volatility is constant. Financial data do not behave that way: volatility clusters, changes over time, and often moves against the asset price. Stochastic volatility models make the volatility itself random.

Prerequisites for Heston-Type Models. Before the Heston model makes sense, keep the following objects straight:

- A Brownian motion models continuous random noise.
- An SDE describes a quantity whose infinitesimal change has a drift part and a random noise part.
- A filtration records the information available up to time t .
- An adapted process cannot use future information.
- Quadratic variation explains why Ito's formula has a second-order correction.
- In option pricing, the risk-neutral measure is the probability measure under which the discounted asset price is a martingale.
- Volatility is the standard deviation scale, while variance is its square.

Definition 6.39: Correlated Brownian Motions.

Two Brownian motions W^S and W^V have correlation $\rho \in [-1, 1]$ if their quadratic covariation satisfies

$$d\langle W^S, W^V \rangle_t = \rho dt.$$

Informally, their small random shocks satisfy

$$dW_t^S dW_t^V = \rho dt.$$

Remark 6.40: Analysis Lens. Correlation of Brownian motions is encoded by quadratic covariation. In finite-dimensional linear algebra, correlation is an inner-product idea. Here the same idea appears infinitesimally through $d\langle W^S, W^V \rangle_t$.

Remark 6.41: Why Correlation Matters. In equity markets, prices and volatility often move in opposite directions: when the stock price falls, volatility often rises. The parameter ρ models this leverage effect. A typical equity model has

$$\rho < 0.$$

Definition 6.42: The Heston Model.

Under a risk-neutral probability measure, the Heston model is

$$\begin{aligned} dS_t &= rS_t dt + \sqrt{V_t} S_t dW_t^S, \\ dV_t &= \kappa(\theta - V_t) dt + \xi \sqrt{V_t} dW_t^V, \\ d\langle W^S, W^V \rangle_t &= \rho dt. \end{aligned}$$

Here S_t is the asset price and V_t is the instantaneous variance. The parameters have the following meanings:

r	risk-free interest rate
κ	speed of mean reversion of variance
θ	long-run variance level
ξ	volatility of volatility
ρ	correlation between price noise and variance noise

Remark 6.43: Structural Lens. Heston is a coupled system of SDEs. The state variable is

$$(S_t, V_t),$$

so the model lives in a two-dimensional state space. The functional analytic object behind pricing is often an operator or semigroup acting on payoff functions. This is why Heston connects stochastic calculus, PDE, Fourier analysis, and operator thinking.

Remark 6.44: How to Read the Heston System. The first equation says that the price behaves like geometric Brownian motion, except that the constant volatility σ has been replaced by the random volatility $\sqrt{V_t}$:

$$\sigma \rightsquigarrow \sqrt{V_t}.$$

The second equation says that variance is pulled back toward θ . If $V_t > \theta$, then $\kappa(\theta - V_t) < 0$ and the drift pushes variance down. If $V_t < \theta$, the drift pushes variance up.

Definition 6.45: CIR Variance Process.

The variance process in the Heston model,

$$dV_t = \kappa(\theta - V_t) dt + \xi \sqrt{V_t} dW_t^V,$$

is a Cox–Ingersoll–Ross, or CIR, process. The square-root noise term

$$\xi \sqrt{V_t} dW_t^V$$

is designed to keep the variance nonnegative.

Remark 6.46: Analysis Lens. The CIR process is a diffusion with state constraint $V_t \geq 0$. The coefficient $\sqrt{V_t}$ degenerates at the boundary, so ordinary Lipschitz intuition is not the whole story. This is a good place to remember that SDE theory is analysis of coefficients, domains, and boundary behavior.

Remark 6.47: Feller Condition. The condition

$$2\kappa\theta \geq \xi^2$$

is called the Feller condition. When it holds, the CIR variance process stays strictly positive if $V_0 > 0$. If it fails, the variance can touch 0, but the square-root structure still prevents it from becoming negative in the usual Heston setup.

Example 6.48: What Heston Adds Beyond Black–Scholes. In Black–Scholes,

$$dS_t = rS_t dt + \sigma S_t dW_t,$$

the volatility σ is constant. Therefore every option price is driven by one volatility number. In Heston, volatility is random:

$$\sigma_t = \sqrt{V_t}.$$

This gives the model more room to fit market features such as volatility smiles, volatility clustering, and the leverage effect. ■

Remark 6.49: Why Heston Is Tractable. Heston is popular because it is a stochastic volatility model that still has strong analytic structure. It is an *affine* model: roughly, the logarithm of the characteristic function of $\log S_t$ can be computed from Riccati-type ordinary differential equations. This is why Fourier pricing methods can be used efficiently.

Definition 6.50: Rough Volatility Idea.

A volatility process is called *rough* when its paths are less regular than Brownian-looking paths at small time scales. A common roughness parameter is a Hurst index

$$H \in (0, \tfrac{1}{2}).$$

Smaller H means rougher local behavior.

Remark 6.51: Why Rough Volatility Appears. Empirically, volatility does not look like a smooth Markov diffusion at very small time scales. It has memory and irregular local behavior. Rough volatility models try to capture this by replacing the Markovian variance equation with a Volterra equation, where the past enters through a singular kernel.

Definition 6.52: Volterra Kernel.

For $\alpha \in (\frac{1}{2}, 1)$, the kernel

$$K(t) = \frac{t^{\alpha-1}}{\Gamma(\alpha)}$$

has a weak singularity at $t = 0$. A Volterra integral of the form

$$\int_0^t K(t-s)f(s) ds$$

weights the past values $f(s)$ according to how close s is to the present time t .

Remark 6.53: Functional-Analysis Lens. A Volterra kernel defines an integral operator:

$$(K * f)(t) = \int_0^t K(t-s)f(s) ds.$$

So rough volatility is not just “more random volatility.” It changes the model from a Markovian SDE into an equation with memory, where the past enters through an operator.

Definition 6.54: Rough Heston Model.

A common rough Heston formulation is

$$\begin{aligned} dS_t &= rS_t dt + \sqrt{V_t} S_t dW_t^S, \\ V_t &= V_0 + \int_0^t K(t-s)\kappa(\theta - V_s) ds + \int_0^t K(t-s)\xi\sqrt{V_s} dW_s^V, \end{aligned}$$

with

$$K(t) = \frac{t^{\alpha-1}}{\Gamma(\alpha)}, \quad \alpha = H + \frac{1}{2}, \quad H \in (0, \frac{1}{2}).$$

As before,

$$d\langle W^S, W^V \rangle_t = \rho dt.$$

Remark 6.55: Structural Lens. Classical Heston is local in time: the next move depends on the current state. Rough Heston is history-dependent: the variance is produced by a Volterra operator applied to the past. Structurally, this is the jump from finite-dimensional Markov state variables to path-dependent, function-space dynamics.

Remark 6.56: Heston Versus Rough Heston. The Heston variance process is Markovian:

$$dV_t = \kappa(\theta - V_t) dt + \xi\sqrt{V_t} dW_t^V.$$

Its future depends on the present value V_t .

Rough Heston is non-Markovian in V_t alone:

$$V_t = V_0 + \int_0^t K(t-s)(\dots) ds + \int_0^t K(t-s)(\dots) dW_s^V.$$

The current variance remembers the whole past through the kernel $K(t-s)$.

Takeaway. The conceptual path is:

Brownian motion $\rightarrow$ Ito integral $\rightarrow$ SDEs
 $\rightarrow$ geometric Brownian motion $\rightarrow$ random variance V_t
 $\rightarrow$ Heston
 $\rightarrow$ Volterra memory kernel $\rightarrow$ rough Heston.

Heston makes volatility random. Rough Heston makes volatility random and locally rough, with memory.

Takeaway. The route into stochastic calculus is:

measure space $\rightarrow$ probability space $\rightarrow$ random variables
 $\rightarrow$ stochastic processes $X(t, \omega) \rightarrow$ adapted processes $\rightarrow$ Brownian motion
 $\rightarrow$ Ito integral $\rightarrow$ Ito formula and SDEs.

Probability becomes analysis on random paths.

CHAPTER 7

PDE Analysis: Elliptic PDE

First Topics

1. Laplace and Poisson equations.
2. Weak derivatives and Sobolev spaces.
3. Integration by parts and weak formulations.
4. Lax–Milgram theorem.
5. Dirichlet and Neumann boundary value problems.
6. Maximum principles.
7. Interior and boundary regularity.

Learning Goals. This section is for elliptic partial differential equations, especially weak solutions, Sobolev spaces, boundary value problems, regularity, and variational methods.

Core Questions

- What does it mean to solve a PDE weakly?
 - How do boundary conditions enter the function space?
 - Why are elliptic equations connected to minimization problems?
 - What regularity can be recovered from weak data?
-

Foundations to Strengthen.

- Know weak derivatives and Sobolev spaces, especially H^1 and H_0^1 .
- Practice integration by parts and Green's identities.
- Translate classical boundary value problems into weak formulations.
- Learn Lax–Milgram as the basic existence theorem.
- Use Young's inequality to split product terms in energy estimates, then combine it with compactness and maximum principles.

Example 7.1. The Poisson problem

$$-\Delta u = f \quad \text{in } \Omega, \quad u = 0 \quad \text{on } \partial\Omega$$

is often studied by replacing classical derivatives with weak derivatives and searching for u in a Sobolev space. ■

Checkpoint. Why is the weak formulation of an elliptic PDE often easier to solve than the classical formulation?

Takeaway. Elliptic PDE analysis turns differential equations into problems about function spaces, estimates, and compactness.

CHAPTER 8

Applications

First Topics

1. Heat flow, diffusion, and energy methods.
2. Elasticity and variational principles.
3. Spectral methods and eigenvalue problems.
4. Probability in function spaces.
5. Optimization in Hilbert and Banach spaces.
6. Statistical learning theory and reproducing kernels.
7. Time series, covariance operators, and spectral ideas.
8. Quantum mechanics as Hilbert-space operator theory.
9. Numerical approximation of PDE.

Learning Goals. This section is for examples where the theory is used: physics, geometry, probability, optimization, numerical methods, data science, machine learning, and models involving PDE or functional analytic tools.

Core Questions

- Which mathematical structures appear naturally in applications?
 - What assumptions make a model analytically tractable?
 - How do estimates translate into stability or robustness?
 - Which parts of the theory are actually used in computation?
-

Foundations to Strengthen.

- In optimization, know convexity, gradients, Hessians, constraints, and KKT conditions.
- In machine learning, separate approximation error, estimation error, regularization, and generalization.
- In RKHS theory, remember: kernels encode inner products and make point evaluation continuous.
- In time series, know stationarity, autocovariance, ARMA models, and spectral ideas.
- In quantum mechanics, focus on Hilbert spaces, self-adjoint operators, spectra, and unitary evolution.

How to Read This Chapter Through Functional Analysis. The fastest way to organize an application is to ask four questions:

1. What is the space? Are the objects vectors, functions, random variables, signals, or wave functions?
2. What is the norm or inner product? What does “small error” mean in this model?
3. What are the maps? Are they operators, functionals, projections, kernels, or semigroups?
4. What theorem is being used? Boundedness, continuity, compactness, projection, spectral theory, or duality?

This chapter is not a list of unrelated applications. It is a set of examples showing the same pattern:

choose a space $\longrightarrow$ put structure on it $\longrightarrow$ study maps on that space.

Application	Functional-analysis object	What it explains
Machine learning	function class with a norm	generalization and regularization
RKHS	Hilbert space of functions	point evaluation as inner product
Time series	L^2 random variables and shifts	prediction, covariance, filters
Quantum mechanics	Hilbert space and operators	states, observables, evolution
PDE models	function spaces and weak derivatives	existence, stability, energy estimates

8.1 Theory of Machine Learning

Big Idea. Machine learning becomes mathematical when we stop thinking only about algorithms and ask what kind of function is being learned, from what data, under what loss, and with what control on complexity. Analysis enters because learning is usually an infinite-dimensional approximation problem: choose a function from a large class, fit finite data, and still hope it generalizes.

Definition 8.1: Learning Problem.

Let (X, Y) be random variables with values in an input space $\mathcal{X}$ and an output space $\mathcal{Y}$. A hypothesis is a function $f : \mathcal{X} \rightarrow \mathcal{Y}$. Given a loss function $\ell(f(x), y)$, the population risk is

$$R(f) = \mathbb{E}[\ell(f(X), Y)].$$

From data $(x_i, y_i)_{i=1}^n$, the empirical risk is

$$R_n(f) = \frac{1}{n} \sum_{i=1}^n \ell(f(x_i), y_i).$$

Learning means choosing f so that $R(f)$ is small, even though only $R_n(f)$ is visible.

Remark 8.2: Why Analysis Appears. The central difficulty is not merely fitting the data. If the function class is too large, one can fit noise. If it is too small, one cannot represent the pattern. Thus learning theory asks for estimates of the form

$$R(f) \leq R_n(f) + \text{complexity penalty} + \text{probability error}.$$

This is why norms, compactness, continuity, concentration, and Hilbert space geometry show up naturally.

Remark 8.3: Functional-Analysis Lens. The functional-analysis object is the hypothesis space $\mathcal{F}$. A learning algorithm is not just a computational recipe; it is choosing one point $f \in \mathcal{F}$. The norm or penalty $\Omega(f)$ says which functions are considered simple. Regularization is therefore a way of putting geometry on the space of possible functions.

Example 8.4: Regularized Empirical Risk. A common learning rule chooses

$$f_n \in \arg \min_{f \in \mathcal{F}} \left\{ \frac{1}{n} \sum_{i=1}^n \ell(f(x_i), y_i) + \lambda \Omega(f) \right\},$$

where $\Omega(f)$ measures complexity and $\lambda > 0$ controls how much we punish complicated functions. In linear regression, $\Omega(f)$ may be a Euclidean norm. In kernel methods, it is often a Hilbert-space norm $\|f\|_{\mathcal{H}}^2$. ■

Takeaway. Machine learning theory is the study of approximation plus probability: fit the observed sample, but control the function class so the fitted function behaves well away from the sample.

8.2 Reproducing Kernel Hilbert Spaces

Big Idea. A reproducing kernel Hilbert space turns evaluation of a function into an inner product. This is powerful because it lets us do nonlinear function learning with Hilbert-space

geometry.

Definition 8.5: Positive Definite Kernel.

A function $K : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ is a positive definite kernel if it is symmetric and, for every finite set $x_1, \dots, x_n \in \mathcal{X}$ and scalars $c_1, \dots, c_n \in \mathbb{R}$,

$$\sum_{i=1}^n \sum_{j=1}^n c_i c_j K(x_i, x_j) \geq 0.$$

Equivalently, the Gram matrix

$$(K(x_i, x_j))_{i,j=1}^n$$

is positive semidefinite for every finite sample.

Definition 8.6: Reproducing Kernel Hilbert Space.

A Hilbert space $\mathcal{H}$ of functions $f : \mathcal{X} \rightarrow \mathbb{R}$ is a reproducing kernel Hilbert space, or RKHS, if for every $x \in \mathcal{X}$ the evaluation map

$$E_x : \mathcal{H} \rightarrow \mathbb{R}, \quad E_x(f) = f(x),$$

is a bounded linear functional. By the Riesz representation theorem, there exists a unique element $K_x \in \mathcal{H}$ such that

$$f(x) = \langle f, K_x \rangle_{\mathcal{H}} \quad \text{for every } f \in \mathcal{H}.$$

The reproducing kernel is

$$K(x, t) = K_t(x) = \langle K_t, K_x \rangle_{\mathcal{H}}.$$

Remark 8.7: Why “Reproducing”? The formula

$$f(x) = \langle f, K_x \rangle_{\mathcal{H}}$$

recovers, or reproduces, the value of the function at x using only the inner product with the kernel section K_x . Point evaluation becomes a geometric operation.

Example 8.8: Kernel Ridge Regression. Given data $(x_i, y_i)_{i=1}^n$ and an RKHS $\mathcal{H}$, kernel ridge regression solves

$$\min_{f \in \mathcal{H}} \left\{ \frac{1}{n} \sum_{i=1}^n (f(x_i) - y_i)^2 + \lambda \|f\|_{\mathcal{H}}^2 \right\}.$$

The representer theorem says that the minimizer has the finite form

$$f_n(x) = \sum_{i=1}^n \alpha_i K(x, x_i).$$

So even though $\mathcal{H}$ may be infinite dimensional, the learned function lives in the span of the kernel sections centered at the data. ■

Remark 8.9: The Functional-Analytic Meaning. The kernel method works because the data only ask for point evaluations $f(x_i)$. In an RKHS, those evaluations are bounded linear functionals, so Riesz turns them into inner products with K_{x_i} . Thus a learning problem over functions becomes linear algebra with the Gram matrix

$$K_{ij} = K(x_i, x_j).$$

This is the bridge from functional analysis to practical machine learning.

Remark 8.10: Structural Translation. In ordinary regression, one often thinks of the unknown as a formula. In an RKHS, the unknown is a vector $f \in \mathcal{H}$. The data impose linear measurements

$$E_{x_i}(f) = f(x_i).$$

The key theorem is Riesz representation: each bounded functional E_{x_i} is represented by a vector K_{x_i} . That is why kernels turn function fitting into Hilbert-space linear algebra.

Takeaway. An RKHS is a function space where evaluation is continuous. That single analytic fact is what makes kernels usable for learning: nonlinear function fitting is converted into Hilbert-space geometry and finite Gram-matrix computation.

8.3 Time Series Analysis

Big Idea. A time series is not only a list of numbers. It is a signal indexed by time, and many of the main questions are questions about operators: shifts, projections, covariance, filters, and spectral decompositions. Functional analysis becomes useful when we treat random variables, paths, or signals as points in a function space.

Definition 8.11: Time Series.

A discrete-time time series is a sequence of random variables

$$(X_t)_{t \in \mathbb{Z}}.$$

It is weakly stationary if

$$\mathbb{E}[X_t] = \mu \quad \text{and} \quad \text{Cov}(X_t, X_{t+h}) = \gamma(h)$$

depend only on the lag h , not on the absolute time t .

Remark 8.12: The Hilbert-Space View. If $X_t \in L^2(\Omega)$, then each observation is a vector in the Hilbert space of square-integrable random variables. The inner product is

$$\langle X, Y \rangle_{L^2} = \mathbb{E}[XY],$$

or, after centering,

$$\langle X - \mathbb{E}X, Y - \mathbb{E}Y \rangle_{L^2} = \text{Cov}(X, Y).$$

Thus covariance is an inner product after removing means. This is why projection, orthogonality, and least-squares prediction appear so naturally in time series.

Remark 8.13: Functional-Analysis Lens. For time series, the space is often $L^2(\Omega)$, the Hilbert space of random variables with finite second moment. The main maps are the shift operator, covariance operator, and projection onto the span of past observations. In this language, forecasting is not magic: it is often an orthogonal projection problem.

Definition 8.14: Backshift Operator.

The backshift operator B acts on a time series by

$$(BX)_t = X_{t-1}.$$

For example, an autoregressive model of order one can be written

$$X_t = \phi X_{t-1} + \varepsilon_t, \quad \text{or} \quad (I - \phi B)X_t = \varepsilon_t.$$

So an AR model is an operator equation: invert a polynomial in the shift operator.

Example 8.15: AR(1) as an Operator Equation. For

$$X_t = \phi X_{t-1} + \varepsilon_t, \quad |\phi| < 1,$$

we formally invert $I - \phi B$:

$$X_t = (I - \phi B)^{-1} \varepsilon_t = \sum_{k=0}^{\infty} \phi^k \varepsilon_{t-k}.$$

The condition $|\phi| < 1$ is exactly the condition that this infinite series behaves well. Stability is an operator-norm idea in disguise. ■

Remark 8.16: Spectral Intuition. Stationary time series also have a frequency-side description. The autocovariance function $\gamma(h)$ and the spectral density $f(\omega)$ are Fourier-transform partners:

$$\gamma(h) = \int_{-\pi}^{\pi} e^{ih\omega} f(\omega) d\omega.$$

This is why complex analysis, Fourier analysis, and operator theory keep returning in time series: they describe the same signal from different coordinates.

Takeaway. Time series analysis becomes less mysterious when viewed as Hilbert-space geometry plus operators. Prediction is projection, covariance is an inner product after centering, ARMA models are equations in the shift operator, and spectral analysis is Fourier analysis of dependence.

8.4 Quantum Mechanics

Big Idea. The mathematical language of quantum mechanics is Hilbert space. States are vectors, observables are operators, measurements are spectral decompositions, and evolution is generated by an operator. This is why functional analysis and operator theory make quantum mechanics feel much more approachable.

Definition 8.17: State Space.

In the Hilbert-space formulation of quantum mechanics, a pure state is a unit vector

$$\psi \in \mathcal{H}, \quad \|\psi\|_{\mathcal{H}} = 1.$$

Two vectors differing by a nonzero complex scalar represent the same physical ray, so the geometry is Hilbert-space geometry up to phase.

Definition 8.18: Observable.

An observable is represented by a self-adjoint operator A on $\mathcal{H}$. If the system is in the state ψ , the expected value of the observable is

$$\langle A\psi, \psi \rangle_{\mathcal{H}}.$$

Self-adjointness matters because it gives real expectation values and allows a spectral interpretation of measurement.

Example 8.19: Position and Momentum. On $\mathcal{H} = L^2(\mathbb{R})$, the position operator is formally

$$(Q\psi)(x) = x\psi(x),$$

and the momentum operator is formally

$$(P\psi)(x) = -i\hbar\psi'(x).$$

They satisfy the commutation relation

$$QP - PQ = i\hbar I$$

on suitable domains. This is one reason domain issues matter: the most important quantum operators are often unbounded, so they cannot be treated like ordinary matrices without care. ■

Definition 8.20: Schrodinger Evolution.

The Schrodinger equation is

$$i\hbar \frac{d}{dt} \psi(t) = H\psi(t),$$

where H is the Hamiltonian operator. Formally, the solution is

$$\psi(t) = e^{-itH/\hbar} \psi(0).$$

Thus time evolution is an operator exponential. When H is self-adjoint, the evolution is unitary, so the norm of the state is preserved.

Remark 8.21: Why Functional Analysis Matters. Finite-dimensional quantum mechanics already looks like linear algebra: vectors, matrices, eigenvalues. Infinite-dimensional quantum mechanics requires functional analysis because wave functions live in spaces such as $L^2(\mathbb{R}^n)$, and the main operators may be unbounded. One must care about domains, adjoints, spectra, convergence, and completeness.

Remark 8.22: Functional-Analysis Lens. The space is the Hilbert space $\mathcal{H}$ of states. A state is a vector, an observable is usually a self-adjoint operator, and time evolution is a one-parameter family of unitary operators. The same three questions keep returning:

What is the space? What is the operator? What is its spectrum?

That is the functional-analysis skeleton behind the physics.

Takeaway. Quantum mechanics is not separate from analysis. It is one of the clearest places where Hilbert spaces, self-adjoint operators, spectral theory, and PDE meet in a single language.

CHAPTER 9

Computational Mathematics

First Topics

1. Numerical analysis.
2. Numerical linear algebra.
3. Numerical differential equations.
4. Numerical solvers.
5. Computational modeling.

Learning Goals. Computational mathematics is where analysis, linear algebra, differential equations, optimization, and modeling become algorithms. The point is not only to compute numbers, but to understand stability, error, convergence, and the assumptions that make computation trustworthy.

Core Questions

- How do continuous mathematical problems become finite algorithms?
 - What kinds of error enter a computation?
 - Why do stability and conditioning matter as much as formulas?
 - How do numerical solvers connect theory with real models?
-

Foundations to Strengthen.

- Separate truncation error, discretization error, roundoff error, and modeling error.
- Learn conditioning versus stability: the problem can be sensitive, or the algorithm can be unstable.
- Make LU, QR, SVD, eigenvalue methods, and least squares routine.
- Know basic ODE solvers, stiffness, and finite difference / finite element ideas for PDE.
- Always ask whether the algorithm converges to the intended mathematical object.

9.1 Numerical Analysis

Big Idea. Numerical analysis studies how to approximate mathematical objects while controlling error. The central question is not just whether an algorithm produces an answer, but whether the answer is stable, accurate, and converges to the object we intended to compute.

Learning Goals.

- Track approximation error, roundoff error, and truncation error.
- Understand convergence rates and asymptotic error estimates.
- Learn why stability is often the difference between useful computation and meaningless output.

Takeaway. Numerical analysis turns proofs into computable procedures, but it also asks whether those procedures deserve to be trusted.

9.2 Numerical Linear Algebra

Big Idea. Numerical linear algebra studies algorithms for linear systems, eigenvalue problems, matrix factorizations, and least-squares problems. Since many nonlinear and differential-equation algorithms reduce to repeated linear algebra steps, this subject is the computational backbone of applied mathematics.

Learning Goals.

- Understand conditioning, matrix norms, and backward error.
- Study Gaussian elimination, QR factorization, SVD, and eigenvalue algorithms.
- Learn why large sparse matrices require iterative methods.

Takeaway. Linear algebra is finite-dimensional, but numerical linear algebra is where finite-dimensional structure meets real computational limits.

9.3 Numerical Differential Equations

Big Idea. Numerical differential equations approximate the evolution or spatial structure described by ODEs and PDEs. The challenge is to replace a continuous problem by a discrete one without destroying the qualitative behavior of the original equation.

Learning Goals.

- Learn time-stepping methods for ODEs, including Euler and Runge–Kutta methods.
- Understand consistency, stability, and convergence.
- Study finite difference, finite element, and spectral approaches to PDEs.

Takeaway. Discretizing a differential equation is not merely sampling it. The discrete problem must preserve enough of the original structure to be meaningful.

9.4 Numerical Solvers

Big Idea. A numerical solver is an algorithmic machine for finding unknowns: roots, fixed points, minimizers, solutions of linear systems, or solutions of differential equations. Solver design is where estimates, stopping criteria, and computational cost become practical decisions.

Learning Goals.

- Compare direct and iterative solvers.
- Understand Newton's method, fixed-point iteration, gradient methods, and Krylov subspace methods.
- Learn how tolerances, residuals, and stopping criteria affect the reliability of a computed answer.

Takeaway. A solver is only as good as its error control. A small residual may be useful, but it must be interpreted through conditioning and model context.

9.5 Computational Modeling

Big Idea. Computational modeling uses mathematics to build a simplified world that can be simulated. A model is not the world itself: it is a structured approximation whose assumptions must be visible.

Learning Goals.

- Identify state variables, parameters, equations, and constraints.
- Separate modeling error from numerical error.
- Understand calibration, validation, sensitivity, and uncertainty.
- Connect mathematical structure with scientific, economic, or engineering interpretation.

Takeaway. Computation does not replace theory. It gives theory a laboratory, and the quality of that laboratory depends on the quality of the model.

CHAPTER 10

Philosophy of Math

First Topics

1. Platonism, formalism, intuitionism, and structuralism.
2. The role of definitions in creating mathematical worlds.
3. Rigor versus intuition.
4. The meaning of existence theorems.
5. The relationship between computation and proof.
6. The unreasonable effectiveness of mathematics.

Learning Goals. This section is for reflecting on what mathematics is doing: abstraction, rigor, proof, intuition, formalism, mathematical objects, and the relationship between mathematics and the world.

Core Questions

- What is a mathematical object?
 - What makes a proof convincing?
 - Why does abstraction make mathematics more powerful?
 - Is mathematics discovered, invented, or both?
 - Why is mathematics so effective in science?
-

Foundations to Strengthen.

- Compare Platonism, formalism, intuitionism, and structuralism without collapsing them into slogans.
 - Ask what definitions do: they create a structure, then consequences become necessary inside that structure.
 - Separate proof, computation, intuition, and explanation.
 - Study existence theorems: what does it mean to prove something exists without constructing it?
 - Connect mathematical explanation with science, not only with calculation.
-

CHAPTER 11

Philosophy of Science and Quant Research

First Topics

1. Models, assumptions, idealization, and approximation.
2. Explanation versus prediction.
3. Probability, uncertainty, and statistical evidence.
4. Backtesting, out-of-sample testing, and model risk.
5. Time series, dependence, stationarity, and regime change.
6. Optimization, transaction costs, and constraints.
7. Functional analysis as a language for signals, spaces, and operators.

Big Idea. Philosophy of science studies how models explain, predict, fail, and get revised. Quant research uses that same discipline under uncertainty: build a model, test it against data, understand its assumptions, and know when its apparent success is only an artifact.

Core Questions

- What is a model, and what does it leave out?
 - What counts as evidence for or against a quantitative strategy?
 - How do prediction, explanation, and causation differ?
 - Why do backtests fail when assumptions, regimes, or data pipelines shift?
 - How can mathematical rigor protect against overfitting and wishful thinking?
-

Foundations to Strengthen.

- Separate model, mechanism, prediction, explanation, and causation.
- Treat uncertainty carefully: sampling error, model risk, regime change, and hidden assumptions are different problems.
- In quant research, know why backtests overfit and why out-of-sample testing matters.
- Learn stationarity, dependence, transaction costs, constraints, and risk controls.
- Ask what evidence would make a model weaker, not only stronger.

Learning Goals. This section is for connecting philosophy of science with quantitative research. A quant model should not be treated as magic. It is a structured claim about the world: what variables matter, how uncertainty is represented, what data can test the claim, and where the model should be expected to break.

Big Idea. Quant research is applied epistemology with mathematics attached. The question is not only “Can this formula fit the past?” but also “What would make me believe this model, what would make me distrust it, and what kind of world would have to exist for the strategy to keep working?”

Intuition / First Principles. A good model is like a controlled simplification. It discards details so that structure becomes visible, but the discarded details can return as risk. Philosophy of science gives names to this tension: idealization, abstraction, confirmation, falsification, robustness, and explanation. Quant research meets the same tension in backtests, noisy data, changing markets, and hidden assumptions.

Remark 11.1: Functional Analysis and Time Series. One reason functional analysis matters for quant research is that time series can be viewed as objects in spaces. A signal can be projected, smoothed, decomposed, approximated, or acted on by an operator. This is why time series analysis can feel unexpectedly approachable after learning functional analysis: data become functions or vectors, filters become operators, and estimation becomes a question about approximation and stability.

Takeaway. The philosophical habit is to ask what a model means. The quant habit is to ask whether it survives contact with data, costs, constraints, and uncertainty. The mature version of the subject needs both.

A Reference and Study Conventions

A.1 Notation

Takeaway. Add symbols here as soon as they first appear in the notes. A good notation table should be short enough to scan but complete enough that future reference does not require hunting through the text.

Symbol	Meaning
$\mathbb{R}$	real numbers
$\mathbb{C}$	complex numbers
$\mathbb{N}$	natural numbers
$\mathbb{F}$	either $\mathbb{R}$ or $\mathbb{C}$
X^*	continuous dual space of X
$\ x\ $	norm of x
$x_n \rightharpoonup x$	weak convergence of x_n to x

A.2 Learning Template

Each lesson should follow this rhythm:

1. State the learning goals.
2. Give the intuitive picture before the formal definition. In each intuition note, include first-principles thinking: start from the primitive objects, name the problem we are trying to solve, and explain why the definition is forced or natural.
3. Write the formal definition or theorem.
4. Work through at least one example.
5. Add a checkpoint question.
6. End with the takeaway.

A.3 Question Log Convention

When a question is answered in these notes, put the answer in the most relevant subject section and mark the date on which the question was asked. If more questions are added to the same section on the same day, continue under the existing date instead of repeating the date.

B Further Reading

B.1 Real Analysis

- Walter Rudin, *Principles of Mathematical Analysis*.
- Terence Tao, *Analysis I* and *Analysis II*.

B.2 Topology

- James R. Munkres, *Topology*.
- John L. Kelley, *General Topology*.

B.3 Measure Theory

- Gerald B. Folland, *Real Analysis*.
- H. L. Royden and P. M. Fitzpatrick, *Real Analysis*.

B.4 Complex Analysis

- Lars Ahlfors, *Complex Analysis*.
- John B. Conway, *Functions of One Complex Variable*.
- Elias M. Stein and Rami Shakarchi, *Complex Analysis*.

B.5 Functional Analysis

- Erwin Kreyszig, *Introductory Functional Analysis with Applications*.
- John B. Conway, *A Course in Functional Analysis*.
- Walter Rudin, *Functional Analysis*.

B.6 Elliptic PDE

- Lawrence C. Evans, *Partial Differential Equations*.
- David Gilbarg and Neil S. Trudinger, *Elliptic Partial Differential Equations of Second Order*.

B.7 Applications and Philosophy

- Richard Courant and David Hilbert, *Methods of Mathematical Physics*.
 - Eugene Wigner, *The Unreasonable Effectiveness of Mathematics in the Natural Sciences*.
 - Stewart Shapiro, *Thinking about Mathematics*.
-